MATH 325: Honours Ordinary Differential Equations

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Abstract

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1 Introduction

Jean-Philippe Lessard (Burnside 1119). Tutorials every wednesday from 9am to 10am, ENGTR 0070, with Eunpyo Bang. Office hours thursday. No textbooks. 25% assignments (2 written assignments 15%, and 5 webworks 10%). 25% Midterm (February 16 - inclass). 50% Final. Since its honours you will deal with analysis. The Midterm is from 4:05 pm to 5:25 pm in MAASS 112, which will include chapters 1, 2, 3 and will be 5 questions long.

2 Prerequisite Knowledge

2.1 Real Analysis

2.1.1 Norms

Definition 1 (Norm)**.** *A norm $\|\cdot\|$ on $\mathbb{R}^n$ is a way of measuring the "size" or "length" of a vector $x \in \mathbb{R}^n$. It must satisfy:*

- $\|x\| \geq 0$, *and* $\|x\| = 0$ *only if* $x = 0$,
- $\|\alpha x\| = |\alpha| \, \|x\|$ *for any scalar* α,
- $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality).*

Example. *The three most common norms on $\mathbb{R}^n$ are:*

- ***Euclidean norm:*** $\|x\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}$ *(standard geometric length),*
- ***Supremum norm:*** $\|x\|_\infty = \max\{|x_1|, \ldots, |x_n|\}$ *(size of the largest component),*
- ℓ^1 ***norm:*** $\|x\|_1 = |x_1| + \cdots + |x_n|$ *(sum of all components, also called the taxicab norm).*

Remark 1. *In this course, any norm can be used unless stated otherwise. The choice of norm does not affect whether a solution exists or is unique.*

2.1.2 Sequences and Convergence

Definition 2 (Cauchy Sequence)**.** *A sequence of vectors $\{x_k\} \subset \mathbb{R}^n$ is called a Cauchy sequence if its terms eventually get arbitrarily close to one another: for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that*

$$\|x_m - x_n\| < \varepsilon \quad \text{for all } m, n \geq N.$$

Theorem 1. *Every Cauchy sequence in $\mathbb{R}^n$ converges to a point in $\mathbb{R}^n$.*

Remark 2. *This is the key fact used in Picard iteration: we show that the sequence of approximate solutions $\{y_k\}$ is Cauchy, and therefore it must converge to an actual solution.*

2.1.3 Uniform Convergence

Definition 3 (Uniform Convergence)**.** *A sequence of functions $f_k : J \to \mathbb{R}^n$ converges* uniformly *to $f : J \to \mathbb{R}^n$ if*

$$\lim_{k\to\infty} \sup_{t\in J} \|f_k(t) - f(t)\| = 0.$$

In plain terms: the approximation error $\|f_k(t) - f(t)\|$ becomes small simultaneously *for all $t \in J$, not just pointwise.*

Theorem 2. *If $f_k \to f$ uniformly on J and each f_k is continuous, then f is also continuous. Moreover,*

$$\lim_{k\to\infty} \int_J f_k(t)\,dt = \int_J f(t)\,dt.$$

Remark 3. *The second statement says that for uniformly convergent sequences, you can swap the limit and the integral. This is used in the Picard iteration proof to pass the limit inside the integral and confirm that the limit function actually solves the IVP.*

2.1.4 The Mean Value Inequality

Theorem 3 (Mean Value Inequality)**.** *Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^n$ be C^1. For any compact set $K \subset U$ and all $x, y \in K$,*

$$\|f(y) - f(x)\| \le \left(\sup_{z\in K} \|Df(z)\|\right) \|y - x\|,$$

where Df denotes the Jacobian matrix of f.

Remark 4. *This generalizes the familiar scalar fact $|f(b) - f(a)| \le \sup |f'| \cdot |b - a|$ to higher dimensions. Its main consequence for this course is that any C^1 function is locally Lipschitz: on a compact set K, the sup of $\|Df\|$ is finite, providing a Lipschitz constant. This is why the hypothesis "f is C^1" is often used as a sufficient condition for the Existence and Uniqueness Theorem.*

3 Intro, Classification, Theorem of Existence & Uniqueness

3.1 Intro

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Definition 4 (Differential Equation)**.** *A differential equation (DE) is a relation that involves an unknown function and some of its derivatives.*

Definition 5 (Ordinary Differential Equation)**.** *An ordinary differential equation (ODE) is a differential equation whose unknown function depends on one independent variable only.*

Definition 6 (Partial Differential Equation)**.** *A partial differential equation (PDE) is a differential equation whose unknown function depends on more than one independent variable.* ***Will not be taught in this course.***

Example (Motion under gravity and linear drag)**.** *A ball of mass m falls subject to gravity and linear air resistance. Let $v(t)$ be its velocity (positive downward). The net force gives*

$$m\frac{dv}{dt} = mg - \lambda v,$$

where $\lambda \geq 0$ is the drag coefficient. Writing $v = y'$, this becomes the second-order ODE

$$my'' + \lambda y' = mg.$$

Example (The Malthusian Growth Model)**.** *Let $N(t)$ be a population size. Assuming the rate of change is proportional to the current population,*

$$\frac{dN}{dt} = rN,$$

where r is the growth rate ($r > 0$) or decay rate ($r < 0$). Separating variables and applying the initial condition $N(0) = N_0$ gives the unique solution

$$N(t) = N_0 e^{rt}.$$

Example (The Logistic Equation)**.** *If the growth rate depends on the population itself, with carrying capacity K, the simplest choice $R(N) = r(1 - N/K)$ gives*

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right),$$

called the logistic equation. It models growth when $N < K$ and decay when $N > K$.

Example (Heat Equation)**.** *The Heat Equation is an example of a PDE. Let $u = u(x,t)$, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$. This PDE denotes the temperature of a body at time t and at position x.*

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3.2 Classification

3.2.1 The Order

Definition 7. *The order of an ODE is the order of the highest derivative that appears in the equation.*

Example. *$N' = rN$ is first order. $y'' + 2y' = e^t$ is second order.*

3.2.2 Scalar vs. System

Definition 8 (Scalar ODE)**.** *A scalar ODE has one unknown function.*

Definition 9 (System of ODEs)**.** *A system of ODEs has several unknown functions.*

Example (Scalar ODE vs System of ODEs)**.** *Scalar:* $y'' + y = 0$.
System: $\begin{cases} y_1' = y_2 \\ y_2' = -y_1 \end{cases}$

Definition 10 (First Order Sytem)**.** *A first order system is an equation of the form*

$$y'(t) = f(y(t),\, t),$$

where the unknown $y(t) = (y_1(t), \ldots, y_n(t)) \in \mathbb{R}^n$ *is a vector-valued function and* $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$. *Written out by component, this is just* n *coupled equations:*

$$y_i'(t) = f_i(y_1, \ldots, y_n,\, t), \qquad i = 1, \ldots, n.$$

Proposition 1 (Reduction of Higher-Order ODEs)**.** *Any single* n*th-order ODE*

$$y^{(n)} = G\Big(t,\, y,\, y', \ldots, y^{(n-1)}\Big)$$

can be recast as a first order system by introducing new variables for each derivative:

$$y_1 = y, \quad y_2 = y', \quad \ldots \quad y_n = y^{(n-1)}.$$

Differentiating each substitution gives the equivalent system

$$\begin{cases} y_1' = y_2, \\ y_2' = y_3, \\ \vdots \\ y_{n-1}' = y_n, \\ y_n' = G(t,\, y_1, \ldots, y_n). \end{cases}$$

The first $n-1$ *equations just say each variable is the derivative of the previous one; the last equation encodes the original ODE.*

Example (System of First Order ODEs)**.** *Consider the second-order ODE*

$$y'' + 2y' + y = e^t.$$

Define new variables

$$y_1 = y, \qquad y_2 = y'.$$

Then the system becomes

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} + \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}, \quad r(t) = \begin{pmatrix} 0 \\ e^t \end{pmatrix}.$$

Problem 1 (System of First Order ODEs)**.** *Rewrite this third order ODE as a first order system:*

$$y''' + 4y' - y = 0$$

3.2.3 Linearity

Definition 11 (Linearity)**.** *An ODE is linear if the unknown function and its derivatives appear only to the first power, are not multiplied together, and are not composed with nonlinear functions. They may be multiplied by functions of the independent variable.*

Example (Linearity)**.** *Consider the following ODEs:*

- *Linear:* $y' + 3y = 0$
- *Linear:* $y'' - 2xy' + y = cos(x)$
- *Non-linear:* $y' + y^2 = 0$
- *Non-linear:* $y'' + sin(y) = 0$

3.2.4 Autonomy

Definition 12 (Autonomy)**.** *The nth order ODE $F(t, y, y', \ldots, y^{(n)}) = 0$ is autonomous if F does not depend explicitly on t, that is, it can be written as $F(y, y', \ldots, y^{(n)}) = 0$. Otherwise, it is non-autonomous.*

Example (Autonomy)**.** *Consider the following ODEs:*

- *Non-autonomous:* $y'' + 2y' + y - e^t = 0$
- *Autonomous:* $N'(t) = rN(t)$
- *Non-autonomous:* $y'(t) = ty(t)$

3.3 Solutions and Initial Value Problems

Definition 13 (Solution)**.** *A solution of $y'(t) = f(y(t), t)$ on an interval J is a differentiable function $y : J \to D$ such that*

$$y'(t) = f(y(t), t) \quad \forall\, t \in J.$$

Example (Explicit Solution)**.** *Consider the following ODE $y' + y = 1$. We can verify that $y(t) = e^{-t} + 1$, and therefore $y'(t) = -e^{-t}$, is a solution on $\mathbb{R}$. Indeed,*

$$y' + y = -e^{-t} + (e^{-t} + 1) = 1.$$

Example (Implicit Solution)**.** *Consider the ODE $y\frac{dy}{dx} = x$. This is a nonautonomous, nonlinear, first-order scalar ODE. Separating $y\,dy = x\,dx$ and integrating gives the implicit solution $x^2 - y^2 = C, \quad C \in \mathbb{R}$.*

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Definition 14 (Initial Value Problem (IVP))**.** *An initial value problem consists of an ODE together with a condition $y(t_0) = y_0$, where the value of the solution is prescribed at an initial time t_0.*

3.4 Existence and Uniqueness Theorem

3.4.1 Lipschitz continuity

Definition 15 (Lipschitz Continuity)**.** *A function $f : D \to \mathbb{R}^n$ is Lipschitz continuous on $D \subseteq \mathbb{R}^n$ if there exists $L \geq 0$ such that*

$$||f(y_1) - f(y_2)|| \leq L||y_1 - y_2|| \quad \forall y_1, y_2 \in D.$$

The smallest such L is called the Lipschitz constant, denoted $\mathrm{Lip}(f)$.

Remark 5 (How to find a Lipschitz constant)**.** *If f is differentiable on $D \subset \mathbb{R}$ and $|f'(y)| \leq M$ for all $y \in D$, then f is Lipschitz with* $\mathrm{Lip}(f) = M$.

Example. *Let $f(y) = 4y - 5$, $D = \mathbb{R}$, and $|| \cdot || = | \cdot |$. Then*

$$|f(y_1) - f(y_2)| = |4y_1 - 4y_2| = 4|y_1 - y_2|.$$

So f is Lipschitz on $\mathbb{R}$ with $Lip(f) = 4$.

Example. *Let*

$$f(y) = \frac{1}{y-1}, \quad f'(y) = -\frac{1}{(y-1)^2}, \quad D = (1, +\infty).$$

This function is not Lipschitz continuous on D since as $y \to 1^+$, the derivative $f'(y)$ approaches $-\infty$, and so the ratio $\frac{|f(y_2)-f(y_1)|}{|y_2-y_1|}$ approaches ∞ for any $y_1, y_2 \in D$. Therefore, there is no Lipschitz constant for f on D.
To make this function Lipschitz continuous, we fix $\delta > 1$ and define $D_\delta = (\delta, +\infty)$. For $y_1, y_2 \in D_\delta$, by the Mean Value Theorem, there exists $z \in (y_1, y_2)$ such that

$$f(y_2) - f(y_1) = f'(z)(y_2 - y_1).$$

So

$$|f(y_2) - f(y_1)| \leq \frac{1}{(z-1)^2}|y_2 - y_1| \leq \frac{1}{(\delta-1)^2}|y_2 - y_1|.$$

Thus f is Lipschitz on D_δ with

$$Lip(f) = \frac{1}{(\delta-1)^2}.$$

3.4.2 Local Lipschitz continuity

Definition 16 (Locally Lipschitz)**.** *Let $D \subseteq \mathbb{R}^n$ be open. A function $f : D \to \mathbb{R}^n$ is* locally Lipschitz *if for every compact set $K \subset D$ there exists $L > 0$ such that*

$$\|f(y_1) - f(y_2)\| \leq L\|y_1 - y_2\| \quad \forall\, y_1, y_2 \in K.$$

Remark 6. *Lipschitz $\Rightarrow$ locally Lipschitz, but not conversely. Any C^1 function is locally Lipschitz (by the Mean Value Inequality from Section 2.1.4).*

Example (Locally Lipschitz but not Lipschitz)**.**

$$f(y) = y^2 \in \mathbb{R}.$$

Since $f'(y) = 2y$ is unbounded on $\mathbb{R}$, no single constant works on the whole domain, so f is not Lipschitz. However, on any bounded set $K = [-M, M]$, $|f'(y)| \leq 2M$, so f is locally Lipschitz.

Example (Lipschitz $\implies$ locally Lipschitz)**.**

$$f(y) = \sin y.$$

Since $|f'(y)| = |\cos y| \leq 1$, $\forall y \in \mathbb{R}$, f is Lipschitz on $\mathbb{R}$, with $Lip(f) = 1$.

Example (Continuous but not locally Lipschitz)**.**

$$f(y) = \sqrt{|y|}.$$

The derivative $f'(y) = \frac{\operatorname{sgn}(y)}{2\sqrt{|y|}} = \begin{cases} \dfrac{1}{2\sqrt{y}}, & y > 0 \\ -\dfrac{1}{2\sqrt{-y}}, & y < 0 \end{cases}$ *is unbounded near $y = 0$, so no Lipschitz constant exists even locally. Hence f is not locally Lipschitz.*

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3.4.3 Integral Form of the IVP

Lemma 4. *A function y solves the IVP if and only if*

$$y(t) = y_0 + \int_{t_0}^{t} f(y(s), s)\, ds.$$

Proof. If $y' = f(y, t)$ and $y(t_0) = y_0$, then by the Fundamental Theorem of Calculus,

$$y(t) = y_0 + \int_{t_0}^{t} f(y(s), s) ds.$$

Conversely, differentiating the right-hand side gives

$$y'(t) = f(y(t), t), \quad y(t_0) = y_0.$$

□

3.4.4 Picard operator

The integral form of the IVP motivates defining the following operator.

Algorithm 1 (Finding an Interval of Existence)**.** *Since $(y_0, t_0) \in D \times (a, b)$ and the set is open, choose $\alpha, \delta > 0$ such that*

$$D_{\alpha,\delta} = \{(y, t) : \|y - y_0\| \leq \alpha,\ |t - t_0| \leq \delta\} \subset D \times (a, b),$$

and set

$$M_{\alpha,\delta} = \sup_{(y,t)\in D_{\alpha,\delta}} \|f(y,t)\| < +\infty, \qquad \varepsilon = \min\left(\delta, \frac{\alpha}{M_{\alpha,\delta}}\right), \qquad J = (t_0-\varepsilon, t_0+\varepsilon).$$ [1]

Definition 17 (Picard Operator)**.** *For any $y : J \to \mathbb{R}^n$ with $y(t_0) = y_0$ and $(y(t), t) \in D_{\alpha,\delta}$ for all $t \in J$, define*

$$T(y)(t) = y_0 + \int_{t_0}^{t} f(y(s), s)\, ds.$$

A function y solves the IVP if and only if $T(y) = y$, i.e. y is a fixed point *of T.*

Lemma 5 (Invariance)**.** *If $y(t_0) = y_0$ and $(y(t), t) \in D_{\alpha,\delta}$ $\forall t \in J$, then $T(y)(t_0) = y_0$ and $(T(y)(t), t) \in D_{\alpha,\delta}$ for all $t \in J$.*[2]

Proof. Clearly $T(y)(t_0) = y_0$. For $t \in J$,

$$\|T(y)(t) - y_0\| \le \int_{t_0}^{t} \|f(y(s), s)\| ds \le M_{\alpha,\delta}|t - t_0| \le M_{\alpha,\delta}\epsilon \le \alpha.$$

Hence $(T(y)(t), t) \in D_{\alpha,\delta}$. □

Example (Picard operator)**.** *Consider the IVP $y' = y$, $y(0) = 1$. It's integral form is $y(t) = 1 + \int_0^t y(s)ds$. Start with the simplest function that satisfies the initial condition: $y_0(t) = 1$. Now iterate:*

$$y_1 = T(y_0) = 1 + \int_0^t 1ds = 1 + t,$$

$$y_2 = T(y_1) = 1 + \int_0^t (1 + s)ds = 1 + t + \frac{t^2}{2},$$

$$y_3 = T(y_2) = 1 + t + \frac{t^2}{2} + \frac{t^3}{6}.$$

Each step adds to the next power of t. We can express this relation as

$$y(t) = \sum_{k=0}^{\infty} \frac{t^k}{k!} = e^t.$$

Indeed, $T(y)(t) = 1 + \int_0^t e^s ds = e^t$, so $T(y) = y$. Therefore, T is an operator that maps a function to a new one, and only the true solution is a fixed point of T.

[1] You will have to memorize this algorithm for the exams when finding an interval of existence.

[2] Basically, If a function stays inside the box, then applying T keeps it inside the box.

3.4.5 Existence and Uniqueness

Theorem 6 (Existence and Uniqueness)**.** *Let $D \subseteq \mathbb{R}^n$ be open and (a, b) an open interval containing t_0. Suppose $f : D \times (a, b) \to \mathbb{R}^n$* [3] *is continuous and locally Lipschitz in y. If $y_0 \in D$, then the IVP*

$$\begin{cases} y' = f(y, t), \\ y(t_0) = y_0 \end{cases}$$

has a unique solution on some open interval J containing t_0.

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To apply the Existence and Uniqueness Theorem, we choose ε so that the solution remains inside the domain $D \times (a, b)$. Consider the IVP

$$\begin{cases} y' = f(y, t), \\ y(t_0) = y_0. \end{cases}$$

The theorem guarantees that there exists a unique solution

$$y : (t_0 - \varepsilon, t_0 + \varepsilon) \to \mathbb{R}^n.$$

Algorithm 2 (Choosing ε)**.** *Choose $\alpha > 0$ and $\delta > 0$ such that*

$$D_{\alpha,\delta} := \{(y, t) : \|y - y_0\| \leq \alpha,\ |t - t_0| \leq \delta\} \subset D \times (a, b).$$

Since $f(y, t)$ is continuous, define

$$M_{\alpha,\delta} = \sup_{(y,t) \in D_{\alpha,\delta}} \|f(y, t)\| < \infty.$$

Then one may take

$$\varepsilon = \min\left(\delta, \frac{\alpha}{M_{\alpha,\delta}}\right).$$

Remark 7 (Maximizing ε)**.** *To determine the largest value of ε for which the Theorem of Existence and Uniqueness ensures the existence of a solution, then one may try to maximize $\min\left(\delta, \frac{\alpha}{M_{\alpha,\delta}}\right)$ by equating both terms together $\delta = \frac{\alpha}{M_{\alpha,\delta}}$.*

Example. *Consider the IVP*

$$y' = t^2 + y^2, \quad y(0) = 0.$$

Determine the largest value of ε for which the Theorem of Existence and Uniqueness ensures the existence of a solution $y : (-\varepsilon, \varepsilon) \to \mathbb{R}$.
First and foremost, define

$$D_{\alpha,\delta} = \{(y, t) : |y - 0| \leq \alpha,\ |t - 0| \leq \delta\} \subset D \times (a, b).$$

[3] $D \times (a, b)$ denotes the set of all pairs (y, t) with $y \in D \subset \mathbb{R}^n$ and $t \in (a, b)$, i.e., all allowed state-time inputs of f.

Set $M_{\alpha,\delta} = \sup_{(y,t)\in D_{\alpha,\delta}} ||f(y,t)||$ *and* $\varepsilon = \min\left(\delta, \frac{\alpha}{M_{\alpha,\delta}}\right)$. *Since* $f(y,t) = t^2 + y^2$ *is* C^1 *on* $\mathbb{R}^2$, *it is locally Lipschitz, so we take* $D = \mathbb{R}$ *and* $(a,b) = \mathbb{R}$. *Now we need to find* $M_{\alpha,\delta}$.

$$M_{\alpha,\delta} = \sup_{(y,t)\in D_{\alpha,\delta}} |t^2 + y^2|,$$

since t^2 *is maximized when* $t = \delta$ *and* y^2 *is maximized when* $y = \alpha$, *then*

$$M_{\alpha,\delta} = \delta^2 + \alpha^2.$$

Now we want to maximize $\varepsilon = \min\left(\delta, \frac{\alpha}{\delta^2+\alpha^2}\right)$ *over* $\alpha, \delta > 0$.

Step 1: Optimize over α. *For fixed* δ, *let* $g(\alpha) = \frac{\alpha}{\delta^2+\alpha^2}$. *Then*

$$g'(\alpha) = \frac{\delta^2 - \alpha^2}{(\delta^2+\alpha^2)^2} = 0 \implies \alpha = \delta,$$

so the best choice is $\alpha = \delta$, *giving* $\varepsilon = \min\left(\delta, \frac{1}{2\delta}\right)$.

Step 2: Optimize over δ. *The minimum* $\min\left(\delta, \frac{1}{2\delta}\right)$ *is maximized when both terms are equal, since below* $\delta = \frac{1}{\sqrt{2}}$ *the first term is the bottleneck, and above it the second term is. Setting* $\delta = \frac{1}{2\delta}$ *gives* $\delta = \frac{1}{\sqrt{2}}$. *Hence the largest guaranteed* ε *is*

$$\varepsilon = \frac{1}{\sqrt{2}}.$$

Problem 2 (Existence and Uniqueness Theorem)**.** *Consider the initial value problem*

$$\begin{cases} y' = t^2 + y^2, \\ y(0) = 0, \end{cases}$$

Determine the largest value of ϵ *for which the Theorem of Existence and Uniqueness ensures the existence of a solution* $y : (-\epsilon, \epsilon) \to \mathbb{R}$.

Problem 3 (LLC)**.** *Consider the initial value problem*

$$\begin{cases} y' = 3y^{2/3}, \\ y(0) = 0. \end{cases}$$

Investigate existence and uniqueness.

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Corollary 6.1 (Picard Iteration & Existence)**.** *Define* $y_0(t) \equiv y_0$ *(constant). For* $k \geq 1$ *set*

$$y_k(t) = T(y_{k-1})(t) = y_0 + \int_{t_0}^{t} f(y_{k-1}(s), s)\, ds.$$

By the invariance lemma, each y_k *stays inside* $D_{\alpha,\delta}$.

Remark 8. *The sequence* $\{y_k\}$ *is Cauchy in* $\mathbb{R}^n$ *(see Appendix 9.1) and therefore converges uniformly to a function* $y : J \to \mathbb{R}^n$ *which satisfies the integral equation, hence solves the IVP.*

3.4.6 Uniqueness

Proof of Uniqueness. Assume $y(t)$ and $z(t)$ are two solutions of the IVP with the same initial condition. By the Lipschitz property of f,

$$\|y(t) - z(t)\| = \| \int_{t_0}^{t} f(y(s), s) - f(z(s), s) ds\| \leq \int_{t_0}^{t} L\|y(s) - z(s)\| ds,$$

for some Lipschitz constant $L > 0$. Define

$$g(t) = \int_{t_0}^{t} \|y(s) - z(s)\| ds,$$

so $g'(t) \leq Lg(t)$ and $g(t_0) = 0$. Grönwall's idea[4] tells us this forces $g(t) = 0$ for all t, so $\|y(t) - z(t)\| = 0$. Hence, $y(t) = z(t)$, meaning two solutions with the same initial condition cannot diverge, this proves the solution is unique. □

4 First-Order Scalar Equation

January 22, 2026

A first-order scalar ODE involves a single unknown function $y(t)$ and its first derivative:

$$y' = f(y, t).$$

We study three main classes: linear, separable, and exact equations.

4.1 First Order Linear Equations

Definition 18 (First Order Linear Equation)**.** *A first-order linear equation has the form*

$$a_0(t)y' + a_1(t)y = g(t), \quad a_0(t) \neq 0.$$

Dividing by $a_0(t)$ gives the standard form

$$y' + P(t)y = Q(t).$$

4.1.1 Integrating Factor Method

To solve $y' + P(t)y = Q(t)$, multiply both sides by a function $\mu(t)$ chosen so that the left-hand side becomes an exact derivative:

$$\mu y' + \mu P(t)y = \frac{d}{dt}(\mu y).$$

This requires $\mu'(t) = \mu(t)P(t)$, which gives

$$\mu(t) = e^{\int P(t)\, dt}.$$

[4] Multiplying by $e^{-L(t-t_0)}$ gives $\frac{d}{dt}(e^{-L(t-t_0)}g(t)) \leq 0$, which shows $e^{-L(t-t_0)}g(t)$ is decreasing and thus $g(t) = 0$

The equation then becomes

$$\frac{d}{dt}(\mu y) = \mu Q(t),$$

and integrating both sides yields

$$y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)Q(t)\,dt + C\right), \quad C \in \mathbb{R}.$$

Example. *Solve* $y' - 2y = 3e^t$ *with* $y(0) = 1$.
Here $P(t) = -2$, $Q(t) = 3e^t$, *so* $\mu(t) = e^{-2t}$. *Multiplying through:*

$$\frac{d}{dt}(e^{-2t}y) = 3e^{-t}.$$

Integrating:

$$e^{-2t}y = -3e^{-t} + C \implies y(t) = -3e^t + Ce^{2t}.$$

Applying $y(0) = 1$ *gives* $C = 4$, *so*

$$y(t) = -3e^t + 4e^{2t}.$$

Remark 9 (Quick Solution Method)**.** *A quick method to solve a first-order linear equation (not presented by the professor) is*

$$y(t) = \frac{\int \mu(t)Q(t)}{\mu(t)}.$$

4.1.2 Existence and Uniqueness for First Order Linear Equations

Theorem 7 (Existence & Uniqueness)**.** *If* P *and* Q *are continuous on an open interval containing* t_0, *then the IVP*

$$y' + P(t)y = Q(t), \quad y(t_0) = y_0$$

has a unique solution on that entire interval.

Remark 10. *Unlike the general E&U theorem from Chapter 3.4.5, this result gives a global solution on the full interval where* P *and* Q *are continuous, not just locally.*

Example. *For the IVP*

$$\begin{cases} y' + \dfrac{1}{t-1}y = \dfrac{1}{\cos t}, \\ y(0) = 1, \end{cases}$$

$P(t)$ *is discontinuous at* $t = 1$ *and* $Q(t)$ *is discontinuous at* $t = \pm\frac{\pi}{2}$. *The largest interval containing* $t_0 = 0$ *on which both are continuous is*

$$J_{\max} = \left(-\frac{\pi}{2}, 1\right).$$

4.2 Separable Equations

January 27, 2026

Definition 19 (Separable Equation)**.** *The ODE* $\frac{dy}{dx} = f(x, y)$ *is* separable *if* $f(x, y) = f_1(x)f_2(y)$ *for some functions* f_1 *and* f_2*.*

Example. $\dfrac{dy}{dx} = -\dfrac{x}{y}$ *is separable with* $f_1(x) = x$*,* $f_2(y) = -\dfrac{1}{y}$*.*
$\dfrac{dy}{dx} = x^2 + y^2$ *is not separable.*

One can rewrite the separable ODE as

$$M(x) + N(y)\frac{dy}{dx} = 0, \tag{1}$$

where $M(x) = -f_1(x)$ and $N(y) = \frac{1}{f_2(y)}$, and letting H_1, H_2 be antiderivatives of M, N respectively, the chain rule gives

$$\frac{d}{dx}(H_1(x) + H_2(y(x))) = 0,$$

so the implicit general solution is

$$H_1(x) + H_2(y) = C, \quad C \in \mathbb{R},$$

or equivalently in integral form,

$$\int \frac{dy}{f_2(y)} = \int f_1(x)\,dx + C.$$

Example (Logistic Equation)**.** *The logistic equation*

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

is separable. Using partial fractions,

$$\int \frac{dN}{N} + \int \frac{dN}{N-K} = \int r\,dt \implies \ln\left|\frac{N}{N-K}\right| = rt + C.$$

Solving with $N(0) = N_0$ *gives*

$$N(t) = \frac{KN_0}{N_0 + (K - N_0)e^{-rt}}.$$

As $t \to \infty$*,* $N(t) \to K$ *(the carrying capacity), provided* $N_0 \neq 0$*.*

4.3 Exact Equations

January 29, 2026

4.3.1 Potential Functions and Integral Curves

Definition 20 (Potential Function)**.** *A function* $\Psi(x, y)$ *is a potential function for the equation* (1) *if*

$$\frac{\partial \Psi}{\partial x} = M \quad \text{and} \quad \frac{\partial \Psi}{\partial y} = N.$$

The general solution is then given implicitly by the level curves

$$\Psi(x, y) = C.$$

Definition 21 (Integral Curve)**.** *An integral curve is the level curve of Ψ passing through a specific initial condition (x_0, y_0):*

$$\Gamma_{(x_0,y_0)} = \{(x, y) \mid \Psi(x, y) = \Psi(x_0, y_0)\}.$$

Remark 11. *For a separable equation, the general solution $\Psi(x, y) = C$ describes all possible solution curves, and picking a particular C gives the integral curve passing through a given point.*

Example. *For $\frac{dy}{dx} = -\frac{x}{y}$, $y \neq 0$, separating gives $y\,dy + x\,dx = 0$, so the potential function is*

$$\Psi(x, y) = \frac{x^2}{2} + \frac{y^2}{2}.$$

The general solution is the family of circles $x^2+y^2 = C$. For $y(0) = 2$, $\Psi(0, 2) = 2$, so the unique integral curve is

$$y(x) = \sqrt{4 - x^2}, \quad J_{\max} = (-2, 2).$$

4.3.2 Definition and Exactness Criterion

Definition 22 (Exact Equation)**.** *The equation (1) is exact on a domain D if there exists a potential function Ψ such that $\Psi_x = M$ and $\Psi_y = N$. Along any solution $y(x)$,*

$$\frac{d}{dx}\Psi(x, y(x)) = 0,$$

so Ψ is constant along solutions.

Theorem 8. *Let $D \subset \mathbb{R}^2$ be open and simply connected*[5]*. Suppose M, N, M_y, N_x are continuous on D. Then (1) is exact on D if and only if*

$$M_y(x, y) = N_x(x, y) \quad \forall\,(x, y) \in D.$$

Proof. ($\Rightarrow$) Suppose it is exact, then $M = \Psi_x$ and $N = \Psi_y$, so $M_y = \Psi_{xy} = \Psi_{yx} = N_x$ by equality of mixed partials.

($\Leftarrow$) If $M_y = N_x$ on a simply connected domain and partials are continuous, one can construct Ψ with $\Psi_x = M$ and $\Psi_y = N$ explicitly by integration.[6] $\square$

Example. *Solve*

$$y \cos x + 2xe^y + (\sin x + x^2 e^y - 1)\,\frac{dy}{dx} = 0.$$

[5] Meaning there are no holes in the domain.

[6] The construction is shown for rectangles in the notes; the general case follows similarly.

Identify $M = y\cos x + 2xe^y$ *and* $N = \sin x + x^2e^y - 1$. *Check:*

$$M_y = \cos x + 2xe^y = N_x. \quad \checkmark$$

So the equation is exact. Integrate $\Psi_x = M$ *in* x*:*

$$\Psi = y\sin x + x^2e^y + c(y).$$

Differentiate in y *and match with* N*:*

$$\Psi_y = \sin x + x^2e^y + c'(y) = \sin x + x^2e^y - 1 \implies c'(y) = -1 \implies c(y) = -y.$$

The general solution is

$$\Psi(x, y) = y\sin x + x^2e^y - y = C.$$

Remark 12. *Every separable equation is exact on a simply connected domain. Writing* $M(x)\,dx + N(y)\,dy = 0$ *gives* $M_y = 0 = N_x$ *automatically, so exactness always holds. The potential function is simply* $\Psi(x, y) = H_1(x) + H_2(y)$.

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4.3.3 Integrating Factors for Non-Exact Equations

If $M_y \neq N_x$, we seek a function $\mu(x, y)$ such that $\mu M\,dx + \mu N\,dy = 0$ is exact. The condition $(\mu M)_y = (\mu N)_x$ expands to

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x.$$

Case: $\mu = \mu(x)$ **only.** Then $\mu_y = 0$ and the condition reduces to

$$\frac{d\mu}{dx} = \mu\,\frac{M_y - N_x}{N}.$$

If $\frac{M_y - N_x}{N}$ depends only on x, this is a separable ODE for $\mu(x)$ that can be solved directly.

Example. *Solve*

$$(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0.$$

Here $M_y = 3x + 2y$ *and* $N_x = 2x + y$*, so* $M_y \neq N_x$*. Compute:*

$$\frac{M_y - N_x}{N} = \frac{x + y}{x(x + y)} = \frac{1}{x},$$

which depends only on x*. Solving* $\frac{d\mu}{dx} = \frac{\mu}{x}$ *gives* $\mu(x) = x$*. Multiplying through by* x*:*

$$(3x^2y + xy^2)\,dx + (x^3 + x^2y)\,dy = 0.$$

Now $M_y = 3x^2 + 2xy = N_x$. $\checkmark$ *Integrating:*

$$\Psi = \int (3x^2y + xy^2)\,dx = x^3y + \frac{x^2y^2}{2} + c(y).$$

Matching $\Psi_y = N$ *gives* $c'(y) = 0$*, so the general solution is*

$$x^3y + \frac{x^2y^2}{2} = C.$$

4.4 Applications

Example (Falling Object with Air Resistance)**.** *An object of mass m falls under gravity with damping $\gamma > 0$. Setting $v(0) = 0$, the equation*

$$v' + \frac{\gamma}{m} v = g$$

has integrating factor $\mu = e^{\frac{\gamma}{m}t}$, giving

$$v(t) = \frac{gm}{\gamma}\left(1 - e^{-\frac{\gamma}{m}t}\right).$$

As $t \to \infty$, the velocity approaches the terminal value $v_\infty = \frac{gm}{\gamma}$.

Example (Mixing Problem)**.** *A tank holds 100 L of brine with y_0 g of salt. Pure salt solution enters at 50 g/L at rate R L/s, and the well-mixed solution exits at the same rate. The salt amount $y(t)$ satisfies*

$$y' + \frac{R}{100} y = 50R, \quad y(0) = y_0.$$

Using integrating factor $\mu = e^{\frac{R}{100}t}$,

$$y(t) = 5000 + (y_0 - 5000)e^{-\frac{R}{100}t}.$$

Regardless of y_0, the salt amount converges to 5000 g as $t \to \infty$.

5 Systems of Linear Equations

February 3, 2026

Many problems involve several unknown functions evolving together. Instead of a single ODE, we study systems of first-order linear ODEs, treating the unknowns as a vector and applying matrix algebra.

We consider systems of the form

$$y'(t) = A(t)y(t) + r(t), \quad t \in Q, \tag{2}$$

where

$$y(t) = \begin{pmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{pmatrix} \in \mathbb{R}^n, \quad A(t) \in M_n(\mathbb{R}), \quad r(t) = \begin{pmatrix} r_1(t) \\ \vdots \\ r_n(t) \end{pmatrix}.$$

5.1 Reduction to First-Order Systems

Any nth-order scalar ODE can be written as a first-order system by introducing new variables for each derivative.

Example. *Convert* $y''' - 2y'' + 4y' - 5y = \sin t + t^3$ *into a first-order system. Set* $y_1 = y$, $y_2 = y'$, $y_3 = y''$:

$$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & -4 & 2 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \sin t + t^3 \end{pmatrix},$$

which is in the form $y' = Ay + r$ *with* $J_{\max} = \mathbb{R}$.

5.2 Existence, Uniqueness, and Solution Structure

Theorem 9 (Existence and Uniqueness)**.** *If every entry of* $A(t)$ *and* $r(t)$ *is continuous on* Q*, then for any* $t_0 \in Q$ *and* $y_0 \in \mathbb{R}^n$*, the IVP*

$$\begin{cases} y'(t) = A(t)y(t) + r(t), \\ y(t_0) = y_0 \end{cases}$$

has a unique solution on all of Q*. Hence* $J_{\max} = Q$.

Definition 23 (Homogeneity)**.** *If* $r(t) = 0$*, the system is homogeneous; otherwise it is non-homogeneous.*

Definition 24 (Control System)**.** *A* linear control system *is a non-homogeneous system of the form*

$$y' = Ay + Bu,$$

where $A \in M_n(\mathbb{R})$ *governs the dynamics,* $u \in \mathbb{R}^m$ *is the control input, and* $B \in M_{n\times m}(\mathbb{R})$ *determines how the control enters the system. The term* Bu *can be thought of as a structured forcing term that one can choose to influence the behavior of the solution.*

We now focus on the homogeneous system

$$y'(t) = A(t)y(t). \tag{3}$$

Definition 25 (Principle of Superposition)**.** *If* $y_1(t), \ldots, y_n(t)$ *are solutions of* (3)*, then any linear combination is also a solution.*

Definition 26 (Linear Independence)**.** *Solutions* $y_1, \ldots, y_n$ *of* (3) *are linearly independent if*

$$c_1 y_1(t) + \cdots + c_n y_n(t) = 0 \ \ \forall\, t \in Q \implies c_1 = \cdots = c_n = 0.$$

Theorem 10 (Solution Space)**.** *The solution set* S *of* (3) *is an* n*-dimensional vector subspace of* $C^1(Q)$.

Proof. **Step 1: Vector space.** If $y_1, y_2 \in S$ and $\alpha_1, \alpha_2 \in \mathbb{R}$,

$$(\alpha_1 y_1 + \alpha_2 y_2)' = \alpha_1 y_1' + \alpha_2 y_2' = A(t)(\alpha_1 y_1 + \alpha_2 y_2),$$

so any linear combination is in S, and the zero function is in S.

Step 2: Basis of dimension n. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{R}^n$, and let y_i be the unique solution of

$$\begin{cases} y' = A(t)y, \\ y(t_0) = e_i \end{cases}$$

for $i = 1, \ldots, n$. By uniqueness, every $z \in S$ is a linear combination of the y_i, so $\{y_1, \ldots, y_n\}$ spans S. Evaluating at t_0 shows they are independent. Hence $\dim S = n$. □

5.2.1 Matrix Solutions and the Wronskian

Definition 27 (Matrix Solution)**.** *A matrix solution of* (3) *is an $n \times n$ matrix whose columns are solutions of* (3)*.*

Proposition 2. *Let $y(t)$ be a matrix solution of* (3)*. Either* $\det(y(t)) \neq 0$ *for all $t \in Q$, or* $\det(y(t)) = 0$ *for all $t \in Q$.*

Proof. Suppose $\det(y(t_0)) = 0$ for some t_0. Then there exists a nonzero $v \in \mathbb{R}^n$ such that $y(t_0)v = 0$. Define $y_1(t) = y(t)v$; since each column of $y(t)$ is a solution, $y_1 \in S$. Also define $y_2(t) = 0 \in S$. Since $y_1(t_0) = 0 = y_2(t_0)$, uniqueness gives $y_1(t) = y_2(t)$ for all t, so $y(t)v = 0$ for all t. Hence the columns of $y(t)$ are linearly dependent and $\det(y(t)) = 0$ for all $t \in Q$. □

Definition 28 (Wronskian)**.** *The Wronskian of $y_1, \ldots, y_n$ is*

$$W(t) = W(y_1, \ldots, y_n)(t) = \det(y(t)).$$

Example. *Let* $y_1(t) = \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix}$, $y_2(t) = \begin{pmatrix} -2\sin 2t \\ -2\cos 2t \end{pmatrix}$.

$$W(t) = \det \begin{pmatrix} \cos 2t & -2\sin 2t \\ -\sin 2t & -2\cos 2t \end{pmatrix} = -2 \neq 0,$$

so y_1 and y_2 are linearly independent.

Definition 29 (Fundamental Matrix Solution)**.** *A matrix solution with nonvanishing Wronskian is called a* fundamental matrix solution*.*

Theorem 11 (Liouville's Formula)**.** *Let $y(t)$ be a matrix solution of* (3) *with $y(t_0) = y_0$. Then*

$$\det(y(t)) = \det(y_0)\, e^{\int_{t_0}^{t} \operatorname{tr}(A(s))\, ds},$$

where $\operatorname{tr}(M)$ *denotes the sum of the diagonal entries of M.*

5.2.2 Recovering the Coefficient Matrix and Solving IVPs

Proposition 3 (Recovering A)**.** *Let $\Phi(t)$ be a fundamental matrix solution of (3). Then $\Phi(t)$ is invertible for all $t \in Q$ and*

$$A(t) = \Phi'(t)\,\Phi(t)^{-1}.$$

Proof. Since $\Phi'(t) = A(t)\Phi(t)$ and $\det \Phi(t) \neq 0$, multiplying on the right by $\Phi(t)^{-1}$ gives the result. □

Example. *Given* $\Phi(t) = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{-t} \end{pmatrix}$,

$$\Phi'(t) = \begin{pmatrix} 2e^{2t} & 0 \\ 0 & -e^{-t} \end{pmatrix}, \quad \Phi(t)^{-1} = \begin{pmatrix} e^{-2t} & 0 \\ 0 & e^{t} \end{pmatrix},$$

so $A = \Phi'(t)\Phi(t)^{-1} = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$.

Since $y_1, \ldots, y_n$ are linearly independent, the general solution is

$$y(t) = c_1 y_1(t) + \cdots + c_n y_n(t).$$

Given $y(t_0) = y_0$, the constants $c = (c_1, \ldots, c_n)^T$ are uniquely determined by

$$\begin{pmatrix} | & & | \\ y_1(t_0) & \cdots & y_n(t_0) \\ | & & | \end{pmatrix} c = y_0 \implies c = \Phi(t_0)^{-1} y_0.$$

5.3 Homogeneous Systems with Constant Coefficients

Consider (3) with $A \in M_n(\mathbb{R})$ constant. We look for solutions of the form

$$y(t) = e^{\lambda t} u, \tag{4}$$

where u is a constant vector. Substituting into (3),

$$e^{\lambda t}(\lambda u) = e^{\lambda t}(Au) \implies Au = \lambda u.$$

So λ must be an eigenvalue of A and u a corresponding eigenvector.

Definition 30 (Kernel)**.** $\ker A = \{u \mid Au = 0\}$ *is the kernel (null space) of A.*

Definition 31 (Characteristic Polynomial, Eigenvalue, Spectrum)**.** *The* characteristic polynomial *of $A \in M_n(\mathbb{R})$ is $p(\lambda) = \det(A - \lambda I)$. An* eigenvalue *is a root of $p(\lambda)$, and the* spectrum *is*

$$\sigma(A) = \{\lambda \mid p(\lambda) = 0\}.$$

5.3.1 Diagonalizable Case: Real Eigenvalues

Definition 32 (Diagonalizable Matrix)**.** *$A \in M_n(\mathbb{R})$ is* diagonalizable *if there exists an invertible P such that $P^{-1}AP$ is diagonal.*

Theorem 12 (Diagonalization)**.** *A is diagonalizable if and only if it has a basis of eigenvectors. In particular, if all eigenvalues of A are real and distinct, then A is diagonalizable.*

Proof. Let $\lambda_1, \ldots, \lambda_n$ be real distinct eigenvalues with eigenvectors $u_1, \ldots, u_n$. Distinct eigenvalues imply the eigenvectors are linearly independent, so they form a basis of $\mathbb{R}^n$. Let $P = [u_1 \cdots u_n]$. Then

$$AP = PD, \quad D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

so $P^{-1}AP = D$. □

Theorem 13 (General Solution via Diagonalization)**.** *If A is diagonalizable with eigenpairs $(\lambda_i, u_i)_{i=1}^n$, the general solution of (3) is*

$$y(t) = \sum_{i=1}^{n} c_i e^{\lambda_i t} u_i.$$

Example (2×2 system)**.** *Let $y' = Ay$ with $A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$.*

Eigenvalues: *$p(\lambda) = (1-\lambda)^2 - 9 = 0$ gives $\lambda_1 = -2$, $\lambda_2 = 4$, so $\sigma(A) = \{-2, 4\}$.*

Eigenvectors: *For $\lambda_1 = -2$: $(A + 2I)u = 0$ gives $u_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. For $\lambda_2 = 4$: $u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.*

Wronskian: *$W(0) = \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} = 2 \neq 0$, so the solutions are independent.*

General solution:

$$y(t) = c_1 e^{-2t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

With *$y(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$:*

$$\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 3/2 \end{pmatrix}.$$

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Example (3×3 system)**.** *Let $y' = Ay$ with $A = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 5 & -2 \\ -3 & 3 & -2 \end{pmatrix}$.*

Eigenvalues: *$p(\lambda) = -(\lambda + 2)(\lambda - 4)^2$, so $\lambda_1 = -2$ and $\lambda_2 = \lambda_3 = 4$.*

Eigenvectors: *For* $\lambda_1 = -2$*:* $(A+2I)u = 0$ *gives* $u_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$*. For* $\lambda = 4$*: row reduction of* $(A-4I)u = 0$ *yields two free variables (*b *and* c *free,* $a = b-2c$*), giving*

$$u_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad u_3 = \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}.$$

Since $\dim\ker(A-4I) = 2 =$ *algebraic multiplicity,* A *is diagonalizable.*

Wronskian: $W(0) = \det \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix} = 6 \neq 0.$

General solution:

$$y(t) = c_1 e^{-2t} \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + c_2 e^{4t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{4t} \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}.$$

Example (Non-diagonalizable)**.** *Let* $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$*. Since* A *is upper triangular,* $\lambda_1 = \lambda_2 = 1$ *(multiplicity 2). Solving* $(A-I)u = 0$ *gives only one independent eigenvector* $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$*, so* A *is* not *diagonalizable.*

5.3.2 Diagonalizable Case: Complex Eigenvalues

If $A \in M_n(\mathbb{R})$ has a complex eigenpair (λ, u) with $\lambda = \alpha + i\beta$ and $u = p + iq$, then a complex solution is

$$y(t) = e^{(\alpha+i\beta)t}(p+iq). \tag{5}$$

Definition 33 (Euler's Formula)**.**

$$e^{ix} = \cos x + i \sin x, \quad x \in \mathbb{R}. \tag{6}$$

Expanding (5) using Euler's formula and separating real and imaginary parts yields two real solutions:

$$y_1(t) = e^{\alpha t}(p\cos\beta t - q\sin\beta t), \qquad y_2(t) = e^{\alpha t}(p\sin\beta t + q\cos\beta t). \tag{7}$$

Proposition 4 (Complex Conjugate Eigenpairs)**.** *Let* $A \in M_n(\mathbb{R})$*. If* (λ, u) *is an eigenpair with* $\lambda \in \mathbb{C}$*, then* $(\bar{\lambda}, \bar{u})$ *is also an eigenpair.*

Proof. From $Au = \lambda u$, taking conjugates and using $\bar{A} = A$ gives $A\bar{u} = \bar{\lambda}\,\bar{u}$. □

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Proposition 5. *If* $\lambda = \alpha + i\beta \in \sigma(A)$ *with* $\beta \neq 0$ *and eigenvector* $u = p + iq$*, then the functions* (7) *are real-valued linearly independent solutions of* (3)*.*

Example. *Let* $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$.

Eigenvalues: $p(\lambda) = (\lambda - 2)^2 + 1 = 0$ *gives* $\lambda_1 = 2 + i$, $\lambda_2 = 2 - i$.

Eigenvector for $\lambda_1 = 2 + i$***:*** $(A - (2 + i)I)u = 0$ *gives* $b = ia$, *so taking* $a = 1$*:*

$$u_1 = \begin{pmatrix} 1 \\ i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{p} + i \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{q}.$$

By Proposition 4, $u_2 = \bar{u}_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix}$.

Problem 4 (Complex eigenvalues)**.** *Let* $\vec{y}' = \begin{pmatrix} -3 & -2 \\ 5 & 3 \end{pmatrix} \vec{y}$.

(a) Find the eigenvalues and eigenvectors.

(b) Find the real-valued solution to the IVP with $y_1(0) = 7$, $y_2(0) = -15$.

5.3.3 Nondiagonalizable Matrices and Generalized Eigenvectors

Remark 13. *If* A *does not admit a basis of eigenvectors, there are not enough solutions of the form* $e^{\lambda t}u$ *to span the solution space.*

Definition 34 (Algebraic and Geometric Multiplicity)**.** *Let* $\lambda_0 \in \sigma(A)$.

1. *The* algebraic multiplicity m *is the largest integer such that* $(\lambda - \lambda_0)^m$ *divides* $p(\lambda)$.

2. *The* geometric multiplicity *is* $\dim \ker(A - \lambda_0 I)$ *(the number of independent eigenvectors for* λ_0*).*

The defect *is* $d_{\lambda_0} = m - \dim \ker(A - \lambda_0 I) \geq 0$. *We call* λ_0 defective *if* $d_{\lambda_0} > 0$.

Example. *Let* $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $p(\lambda) = (\lambda - 1)^2$. *Algebraic multiplicity* $m = 2$*;* $\ker(A - I) = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ *has dimension 1. Defect* $= 1 > 0$, *so* $\lambda_0 = 1$ *is defective.*

Example. *Let* $A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & -2 & -1 & -1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$, $p(\lambda) = \lambda^3(\lambda + 1)$. *For* $\lambda_0 = 0$*: algebraic multiplicity* $m = 3$, $\ker A = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix} \right\}$ *has dimension 1. Defect* $= 2$.

Example. *For* $A = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 5 & -2 \\ -3 & 3 & -2 \end{pmatrix}$, $p(\lambda) = (\lambda+2)(\lambda-4)^2$. *For* $\lambda_0 = 4$*:* $m = 2$, $\dim\ker(A - 4I) = 2$. *Defect* $= 0$*, so* $\lambda_0 = 4$ *is* not *defective.*

Definition 35 (Generalized Eigenvector)**.** *Let* $\lambda \in \sigma(A)$ *with algebraic multiplicity* m*. A* generalized eigenvector *satisfies* $(A - \lambda I)^m u = 0$.

Theorem 14 (Generalized Eigenvector Basis)**.** *If* $\lambda \in \sigma(A)$ *has algebraic multiplicity* m*, then* $\dim\ker(A - \lambda I)^m = m$*. Consequently, every* $A \in M_n(\mathbb{R})$ *has a basis of generalized eigenvectors.*

Denote $k_j = \ker(A - \lambda I)^j$. Then

$$k_0 = \{0\} \subsetneq k_1 \subsetneq k_2 \subsetneq \cdots \subsetneq k_m.$$

Lemma 15. $k_j \subseteq k_{j+1}$ *for all* j*. Moreover, if* $k_j = k_{j+1}$ *for some* j*, then* $k_j = k_{j+1} = \cdots = k_m$.

Proof. The inclusion is immediate. For the second part: assume $k_j = k_{j+1}$ and let $w \in k_{j+2}$. Setting $u = (A - \lambda I)w \in k_{j+1} = k_j$ gives $(A - \lambda I)^{j+1}u = (A - \lambda I)^{j+2}w = 0$, hence $(A - \lambda I)^j u = 0$, so $(A - \lambda I)^{j+1}w = 0$, meaning $w \in k_{j+1}$. Iterate to get the result. □

Algorithm 3 (Computing Generalized Eigenvectors)**.** *Given a basis of* k_i *with* $\dim(k_i) < m$*, find* $u \in k_{i+1} \setminus k_i$ *by solving* $(A - \lambda I)u = w$ *for each basis vector* $w \in k_i$.

Example. *For* $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$*:* $k_1 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$*. To find* $u \in k_2 \setminus k_1$*, solve*

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

This gives $b = 1$*, a free. Pick* $a = 0$*:* $u_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$*. Check:* $(A - I)^2 u_2 = (A - I)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$. ✓

5.3.4 Practical Solution Guide for Repeated Eigenvalues

Let λ be an eigenvalue with algebraic multiplicity 2.

- If $\dim\ker(A - \lambda I) = 2$ (two independent eigenvectors), then
$$y(t) = c_1 e^{\lambda t} v_1 + c_2 e^{\lambda t} v_2.$$
- If $\dim\ker(A - \lambda I) = 1$ (only one eigenvector v), then the system is defective and we must introduce a generalized eigenvector w satisfying
$$(A - \lambda I)w = v.$$
The general solution is
$$y(t) = e^{\lambda t}\left(c_1 v + c_2(tv + w)\right).$$

higher multiplicity

If an eigenvalue λ has multiplicity $m > 2$ and the system is fully defective (only one eigenvector v), then we build a chain of generalized eigenvectors:

$$(A - \lambda I)w_1 = v, \quad (A - \lambda I)w_2 = w_1, \quad \dots$$

The solution follows the pattern:

$$y(t) = e^{\lambda t}\left[c_1 v + c_2(tv + w_1) + c_3\left(\frac{t^2}{2}v + tw_1 + w_2\right) + \cdots\right].$$

Remark 14. *If you have an eigenvalue with multiplicity $m \implies$ highest power is t^{m-1}.*

5.4 The Matrix Exponential

February 17, 2026

Definition 36 (Matrix Exponential)**.** *For $A \in M_n(\mathbb{R})$,*

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \cdots$$

Definition 37 (Induced Matrix Norm)**.**

$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|.$$

Proposition 6 (Norm Properties)**.** *For all $A, B \in M_n(\mathbb{R})$ and $x \in \mathbb{R}^n$:*

$$\|Ax\| \leq \|A\| \cdot \|x\|, \qquad \|AB\| \leq \|A\| \cdot \|B\|.$$

Proof. For any nonzero x: $\|Ax\| = \|A(x/\|x\|)\| \cdot \|x\| \leq \|A\| \cdot \|x\|$. For submultiplicativity, let $\|x\| = 1$: $\|ABx\| \leq \|A\| \cdot \|Bx\| \leq \|A\| \cdot \|B\|$. Take the supremum over unit vectors. □

Proposition 7 (Convergence)**.** *The series defining e^A converges absolutely for every $A \in M_n(\mathbb{R})$.*

Proof. By the triangle inequality, $\| \sum_{k=0}^{N} A^k/k! \| \leq \sum_{k=0}^{N} \|A\|^k/k!$, which converges to $e^{\|A\|}$. □

Proposition 8 (Basic Properties)**.** *For all $A \in M_n(\mathbb{R})$ and $t, s \in \mathbb{R}$:*

$$e^{A \cdot 0} = I, \quad e^{A(t+s)} = e^{At}e^{As}, \quad e^{-At} = (e^{At})^{-1}, \quad \frac{d}{dt}e^{At} = Ae^{At} = e^{At}A.$$

If $AB = BA$, then $e^{A+B} = e^A e^B$.

Proof of the derivative formula.

$$\frac{d}{dt}e^{At} = \frac{d}{dt}\sum_{k=0}^{\infty}\frac{A^k t^k}{k!} = \sum_{k=1}^{\infty}\frac{A^k t^{k-1}}{(k-1)!} = A\sum_{k=0}^{\infty}\frac{A^k t^k}{k!} = Ae^{At}.$$

□

Theorem 16 (Fundamental Matrix Solution)**.** *e^{At} is a fundamental matrix solution of (3). The general solution is $y(t) = e^{At}C$, and the unique solution to the IVP $y(0) = y_0$ is $y(t) = e^{At}y_0$.*

Proof. $\frac{d}{dt}e^{At} = Ae^{At}$ shows each column satisfies the system. $e^{A\cdot 0} = I$ implies $\det(e^{A\cdot 0}) = 1 \neq 0$, so e^{At} is invertible. □

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5.4.1 Computing e^{At} from a Fundamental Matrix

Let $Y(t)$ be any fundamental matrix solution of $y' = Ay$. Since e^{At} is also a fundamental matrix, there exists invertible C such that $e^{At} = Y(t)C$. Evaluating at $t = 0$:

$$I = Y(0)C \implies C = Y(0)^{-1}, \quad \therefore \quad e^{At} = Y(t)\,Y(0)^{-1}.$$

Example. *For $A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}$ with $\lambda_1 = -2$, $u_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\lambda_2 = 4$, $u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$:*

$$Y(t) = \begin{pmatrix} -e^{-2t} & e^{4t} \\ e^{-2t} & e^{4t} \end{pmatrix}, \quad Y(0)^{-1} = \frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix},$$

$$e^{At} = Y(t)Y(0)^{-1} = \begin{pmatrix} \frac{1}{2}(e^{-2t} + e^{4t}) & \frac{1}{2}(-e^{-2t} + e^{4t}) \\ \frac{1}{2}(-e^{-2t} + e^{4t}) & \frac{1}{2}(e^{-2t} + e^{4t}) \end{pmatrix}.$$

Verify: $e^{A\cdot 0} = I$. ✓

5.4.2 Action of e^{At} on Eigenvectors and Generalized Eigenvectors

Eigenvectors. If $Au = \lambda u$, then

$$e^{At}u = \sum_{k=0}^{\infty}\frac{A^k t^k}{k!}u = \sum_{k=0}^{\infty}\frac{\lambda^k t^k}{k!}u = e^{\lambda t}u.$$

Generalized eigenvectors. If $(A - \lambda I)^{j+1}u = 0$, write $A = (A - \lambda I) + \lambda I$. Since these terms commute,

$$e^{At}u = e^{\lambda t}e^{(A-\lambda I)t}u = e^{\lambda t}\sum_{k=0}^{j}\frac{(A - \lambda I)^k t^k}{k!}u,$$

a finite sum since $(A - \lambda I)^k u = 0$ for $k > j$.

Proposition 9. *The general solution of* (3) *is*

$$y(t) = \sum_{i=1}^{n} c_i\, e^{At} u_i, \quad c_i \in \mathbb{R},$$

where $u_1, \ldots, u_n$ *are* n *linearly independent generalized eigenvectors of* A.

Example. *For* $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\sigma(A) = \{1\}$. *From the generalized eigenvector computation above,* $u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ *(eigenvector) and* $u_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ *(generalized). Then*

$$y_1(t) = e^t u_1, \qquad y_2(t) = e^{At} u_2 = e^t (u_2 + t u_1) = e^t \begin{pmatrix} t \\ 1 \end{pmatrix},$$

giving the general solution

$$y(t) = c_1 e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^t \begin{pmatrix} t \\ 1 \end{pmatrix}.$$

Example. *For* $A = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}$: $p(\lambda) = \lambda^2$, *so* $\sigma(A) = \{0\}$ *with multiplicity 2.* $k_1 = \ker A = \left\langle \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\rangle$. *Solve* $Au_2 = u_1$ *for* $u_2 \in k_2 \setminus k_1$: *choosing* $b = 0$ *gives* $u_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. *Then*

$$y_1(t) = \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \qquad y_2(t) = u_2 + t u_1 = \begin{pmatrix} -1 - 2t \\ t \end{pmatrix},$$

$$y(t) = c_1 \begin{pmatrix} -2 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 - 2t \\ t \end{pmatrix}, \quad W(t) = \det \begin{pmatrix} -2 & -1 - 2t \\ 1 & t \end{pmatrix} = 1 \neq 0.$$

5.5 Non-Homogeneous Systems

Consider

$$y'(t) = Ay(t) + r(t), \tag{8}$$

with $r : I \to \mathbb{R}^n$ continuous. If $y_1(t)$ and $y_2(t)$ are two solutions, then $y_2 - y_1$ solves the homogeneous system, so there exists $C \in \mathbb{R}^n$ with $y_2(t) - y_1(t) = Y(t)C$.

Theorem 17 (General Solution)**.** *Let* $y_p(t)$ *be any particular solution of* (8). *Then the general solution is*

$$y(t) = Y(t)C + y_p(t), \quad C \in \mathbb{R}^n.$$

5.5.1 Variation of Parameters

Look for a particular solution $y_p(t) = Y(t)C_p(t)$, where $C_p(t)$ is unknown. Differentiating and substituting into (8):

$$AY(t)C_p + Y(t)C_p' = AY(t)C_p + r(t) \implies Y(t)C_p'(t) = r(t).$$

Since $Y(t)$ is invertible, $C_p'(t) = Y^{-1}(t)r(t)$, giving

$$y_p(t) = Y(t) \int Y^{-1}(t)\, r(t)\, dt.$$

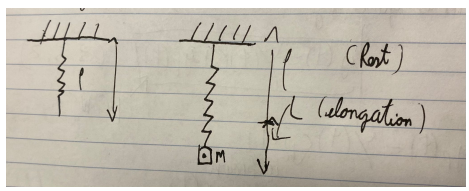
The general solution is therefore

$$y(t) = Y(t)\left[C + \int Y^{-1}(t)\, r(t)\, dt\right], \quad C \in \mathbb{R}^n.$$

5.6 Applications

5.6.1 Mass-Spring Systems

February 24, 2026



At equilibrium, $mg + kL = 0$.

Definition 38 (Hooke's Law)**.** *The restoring spring force is $F_s = -kx$, where $k > 0$ is the spring constant.*

Let $u(t)$ be the displacement from equilibrium ($u > 0$ downward), with damping $F_d = -\gamma u'$ and external force $F(t)$. Newton's second law and the equilibrium relation give

$$mu'' + \gamma u' + ku = F(t). \tag{9}$$

As a first-order system with $y = \begin{pmatrix} u \\ u' \end{pmatrix}$:

$$y' = \begin{pmatrix} 0 & 1 \\ -k/m & -\gamma/m \end{pmatrix} y + \begin{pmatrix} 0 \\ F(t)/m \end{pmatrix}.$$

The characteristic polynomial of the homogeneous part is

$$p(\lambda) = \lambda^2 + \frac{\gamma}{m}\lambda + \frac{k}{m}, \qquad \lambda_\pm = \frac{-\gamma \pm \sqrt{\gamma^2 - 4km}}{2m}.$$

Setting $\Delta = \gamma^2 - 4km$:

- **Overdamped** ($\Delta > 0$): two distinct real negative eigenvalues,

$$u(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t} \to 0 \text{ without oscillation.}$$

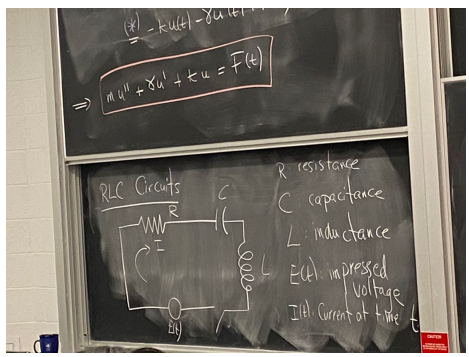
- **Critically damped** ($\Delta = 0$): repeated eigenvalue $\lambda = -\gamma/(2m)$,

$$u(t) = (c_1 + c_2 t)e^{\lambda t} \to 0 \text{ as fast as possible without oscillating.}$$

- **Underdamped** ($\Delta < 0$): complex eigenvalues $\lambda = -\frac{\gamma}{2m} \pm i\omega$ with $\omega = \frac{\sqrt{4km-\gamma^2}}{2m}$,

$$u(t) = e^{-\frac{\gamma}{2m}t}(c_1 \cos \omega t + c_2 \sin \omega t), \text{ exponentially decaying oscillations.}$$

5.6.2 RLC Circuits



Let $Q(t)$ be the capacitor charge and $I(t) = Q'(t)$ the current.

Definition 39 (Kirchhoff's Voltage Law)**.** *The sum of voltage drops around a closed circuit equals the applied voltage $E(t)$.*

The voltage drops are: resistor RI, capacitor Q/C, inductor LI'. Applying Kirchhoff's law and substituting $I = Q'$:

$$LQ'' + RQ' + \frac{1}{C}Q = E(t). \tag{10}$$

This is mathematically identical to (9) under the correspondence

$$m \leftrightarrow L, \quad \gamma \leftrightarrow R, \quad k \leftrightarrow \frac{1}{C}.$$

6 Stability, Phase Portraits and Orbits

March 10, 2026

Let $y'(t) = f(y(t))$, f locally Lipschitz, and $y_0 \in \mathbb{R}^n$. Denote by

$$\Gamma_{y_0} = \{y(t) \mid y'(t) = f(y(t)),\ y(0) = y_0,\ t \in J_{\max}\},$$

the *orbit* of y_0, where $J_{\max}$ is the maximal interval of existence. The orbit is a curve in $\mathbb{R}^n$.

6.1 Orbits and Phase Portraits

Definition 40 (Phase Portrait)**.** *The phase portrait is the collection of all orbits of the system $y' = f(y)$.*

Example. *Let $y' = y(1-y)$. Since this is a separable ODE, the solution is*

$$y(t) = \frac{y_0}{y_0 + (1-y_0)e^{-t}}.$$

The orbits are:

- $\Gamma_0 = \{0\}$,
- $\Gamma_1 = \{1\}$,
- $\Gamma_{y_0} = (0,1)$ *for* $y_0 \in (0,1)$,
- $\Gamma_{y_0} = (1,+\infty)$ *for* $y_0 > 1$,
- $\Gamma_{y_0} = (-\infty,0)$ *for* $y_0 < 0$.

The values 0 *and* 1 *are the equilibrium solutions.*

6.2 Equilibrium Solutions

Definition 41 (Equilibrium Solution)**.** *A vector $\tilde{y} \in \mathbb{R}^n$ is called an* equilibrium solution *(or steady state) of $y' = f(y)$ if $f(\tilde{y}) = 0$. Since $\tilde{y}' = 0 = f(\tilde{y})$, the solution satisfying $y(0) = \tilde{y}$ is constant: $\tilde{y}(t) = \tilde{y}$ for all $t \in \mathbb{R}$, and its orbit is the single point $\Gamma_{\tilde{y}} = \{\tilde{y}\}$.*

Example. *For the linear system $y' = Ay$, equilibrium solutions satisfy $Au = 0$, i.e. they lie in* $\ker(A)$.

- *If A is invertible, then* $\ker(A) = \{0\}$, *so $\tilde{y} = 0$ is the only equilibrium.*
- *If A is not invertible,* $\ker(A)$ *contains more equilibria.*

Example. *Let* $A = \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$. *Solving $Au = 0$ gives $a = b$, so*

$$\ker(A) = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle = \left\{ c \begin{pmatrix} 1 \\ 1 \end{pmatrix} \,\middle|\, c \in \mathbb{R} \right\}.$$

Every point on this line is an equilibrium solution.

Example (Nonlinear Pendulum)**.** *From Newton's second law,*

$$\theta'' + \gamma\theta' + \frac{g}{l}\sin(\theta) = 0.$$

Setting $y_1 = \theta$, $y_2 = \theta'$, *the system becomes*

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} y_2 \\ -\frac{g}{l}\sin(y_1) - \gamma y_2 \end{pmatrix} =: f(y).$$

Equilibria satisfy $f(y) = 0$*: we need* $y_2 = 0$ *and* $\sin(y_1) = 0$*, giving*

$$y_1 = n\pi, \quad n \in \mathbb{Z}.$$

6.3 Stability

6.3.1 Definitions

Definition 42 (Stable Equilibrium)**.** *An equilibrium* $\tilde{y}$ *of* $y' = f(y)$ *is* stable *if for every* $\epsilon > 0$ *there exists* $\delta = \delta(\epsilon) > 0$ *such that every solution with* $\|y(0) - \tilde{y}\| \leq \delta$ *satisfies*

$$\|y(t) - \tilde{y}\| \leq \epsilon \quad \forall\, t \geq 0.$$

Intuitively: solutions that start close to $\tilde{y}$ *remain close for all future time.*

Definition 43 (Unstable Equilibrium)**.** *An equilibrium* $\tilde{y}$ *is* unstable *if it is not stable: there exists* $\epsilon > 0$ *such that for every* $\delta > 0$*, there is a solution* $y(t)$ *with* $\|y(0) - \tilde{y}\| \leq \delta$ *and a time* $T > 0$ *for which* $\|y(T) - \tilde{y}\| > \epsilon$*.*

Definition 44 (Asymptotic Stability)**.** *An equilibrium* $\tilde{y}$ *is* asymptotically stable *if it is stable and there exists* $\alpha > 0$ *such that*

$$\|y(0) - \tilde{y}\| \leq \alpha \implies \lim_{t \to +\infty} \|y(t) - \tilde{y}\| = 0.$$

6.3.2 Examples

March 12, 2026

Example (Stable equilibrium)**.** *Let* $y' = -2y$*. The unique equilibrium is* $\tilde{y} = 0$*, and the solution is* $y(t) = e^{-2t}y(0)$*. For* $t \geq 0$*,*

$$|y(t) - 0| = e^{-2t}|y(0)| \leq |y(0)|.$$

Given $\epsilon > 0$*, choose* $\delta = \epsilon$*. Then* $|y(0)| \leq \delta$ *implies* $|y(t)| \leq \epsilon$ *for all* $t \geq 0$*, so* $\tilde{y} = 0$ *is stable.*

Example (Logistic equation)**.** *Let* $y' = y(1 - y)$*. The equilibria are* $\tilde{y} = 0$ *and* $\tilde{y} = 1$*.*

$\tilde{y} = 0$ ***is unstable.*** *Take* $\epsilon = \frac{1}{2}$ *and any* $0 < \delta < 1$*. Choose* $y(0) = \delta$*; the solution is*

$$y(t) = \frac{\delta}{\delta + (1 - \delta)e^{-t}}.$$

There exists $T = T(\delta)$ such that $y(T) \geq \frac{1}{2}$, so $|y(T) - 0| \geq \epsilon$. Hence $\tilde{y} = 0$ is unstable.

$\tilde{y} = 1$ ***is asymptotically stable.*** *For any $y_0 > 0$,*

$$|y(t) - 1| = \left| \frac{(y_0 - 1)e^{-t}}{y_0 + (1 - y_0)e^{-t}} \right| = \frac{|y_0 - 1|}{y_0 e^t + (1 - y_0)} \leq |y_0 - 1|,$$

since $y_0(e^t - 1) + 1 \geq 1$ for $t \geq 0$. Choosing $\delta = \epsilon$ gives stability. Moreover, as $t \to +\infty$ the numerator $|y_0 - 1|e^{-t} \to 0$, so $y(t) \to 1$ and $\tilde{y} = 1$ is asymptotically stable.

Example (Linear system)**.** *Let $y' = Ay$ with*

$$A = \begin{pmatrix} -\frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{3}{2} \end{pmatrix}.$$

Since $\det(A) \neq 0$, the only equilibrium is $\tilde{y} = 0$.

A fundamental matrix solution is

$$Y(t) = \begin{pmatrix} -e^{-2t} & e^{-t} \\ e^{-2t} & e^{-t} \end{pmatrix},$$

giving

$$e^{At} = Y(t)Y(0)^{-1} = \frac{1}{2} \begin{pmatrix} e^{-2t} + e^{-t} & -e^{-2t} + e^{-t} \\ -e^{-2t} + e^{-t} & e^{-2t} + e^{-t} \end{pmatrix}.$$

Using the induced sup-norm $\|x\|_\infty = \max\{|x_1|, |x_2|\}$ and the corresponding matrix norm

$$\|M\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^{n} |M_{ij}|,$$

one checks that $\|e^{At}\|_\infty \leq 2e^{-t}$. Therefore

$$\|y(t)\|_\infty = \|e^{At} y(0)\|_\infty \leq 2e^{-t} \|y(0)\|_\infty.$$

Stability: *given $\epsilon > 0$, choose $\delta = \frac{\epsilon}{2}$. Then $\|y(0)\|_\infty \leq \delta$ implies $\|y(t)\|_\infty \leq 2e^{-t}\delta \leq \epsilon$ for all $t \geq 0$.*

Asymptotic stability: *for any $\alpha > 0$, if $\|y(0)\|_\infty \leq \alpha$ then $\|y(t)\|_\infty \leq 2\alpha e^{-t} \to 0$ as $t \to +\infty$. Hence $\tilde{y} = 0$ is asymptotically stable.*

6.4 Phase Portraits of Planar Linear Systems

March 10, 2026

Let (3) hold with $A \in M_2(\mathbb{R})$ and $\det(A) \neq 0$. We classify the phase portrait by the eigenstructure of A.

6.4.1 Real Eigenvalues: Diagonalizable Case

The general solution is

$$y(t) = c_1 e^{\lambda_1 t} u_1 + c_2 e^{\lambda_2 t} u_2.$$

March 17, 2026

6.4.2 Complex Eigenvalues

Suppose A has complex eigenvalues $\lambda = \alpha + i\beta$ with $\beta \neq 0$. From Chapter 3, the general solution is

$$y(t) = c_1 e^{\alpha t}(\cos(\beta t)\, p - \sin(\beta t)\, q) + c_2 e^{\alpha t}(\sin(\beta t)\, p + \cos(\beta t)\, q),$$

where $p = \mathrm{Re}(u)$ and $q = \mathrm{Im}(u)$. Given constants c_1, c_2, there exist $v_1, v_2 \in \mathbb{R}^2$ perpendicular such that

$$y(t) = e^{\alpha t}\cos(\beta t)\, v_1 + e^{\alpha t}\sin(\beta t)\, v_2.$$

The nature of the equilibrium $\tilde{y} = 0$ depends on α:

- $\alpha = 0$: $y(t) = \cos(\beta t)v_1 + \sin(\beta t)v_2$ traces a closed ellipse. The origin is called a *stable center*.
- $\alpha > 0$: orbits spiral outward. The origin is called an *unstable spiral point*.
- $\alpha < 0$: orbits spiral inward. The origin is called an *asymptotically stable spiral point*.

Example. *Let* $y' = \begin{pmatrix} -1 & 6 \\ -3/2 & -1 \end{pmatrix} y$ *with* $\sigma(A) = \{-1 \pm 3i\}$. *Since* $\alpha = -1 < 0$, *the origin is an asymptotically stable spiral point. Note that*

$$A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -3/2 \end{pmatrix}.$$

6.4.3 Repeated Eigenvalue: Non-Diagonalizable Case

If $\det(A) \neq 0$, $\sigma(A) = \{\lambda, \lambda\}$, and $\ker(A - \lambda I) = \langle u \rangle$ (only one independent eigenvector), let v be the generalized eigenvector satisfying $(A - \lambda I)v = u$. Then the general solution is

$$y(t) = C_1 e^{\lambda t} u + C_2 e^{\lambda t}(v + tu).$$

Example (Love Affairs)**.** *Let* $R(t)$ *denote the love* (> 0) *or hate* (< 0) *that Romeo feels toward Juliet, and* $J(t)$ *the same for Juliet toward Romeo. In this story, Romeo echoes Juliet's feelings (the more she loves/hates him, the more he loves/hates her), while Juliet reacts negatively to Romeo's interest. This gives*

$$\frac{dR}{dt} = \alpha J, \quad \alpha > 0, \qquad \frac{dJ}{dt} = -\beta R, \quad \beta > 0,$$

or in matrix form,

$$\begin{pmatrix} R' \\ J' \end{pmatrix} = \begin{pmatrix} 0 & \alpha \\ -\beta & 0 \end{pmatrix}\begin{pmatrix} R \\ J \end{pmatrix}.$$

The characteristic polynomial is $p(\lambda) = \lambda^2 + \alpha\beta = 0$, *giving* $\lambda = \pm i\sqrt{\alpha\beta}$. *Since* $\alpha = 0$, *the origin is a* stable center*: their relationship oscillates forever without settling.*

Extended model. *Now suppose both Romeo and Juliet echo each other's feelings equally and are equally cautious. With $a < 0 < b$:*

$$A = \begin{pmatrix} a & b \\ b & a \end{pmatrix}, \qquad p(\lambda) = (a-\lambda)^2 - b^2 = 0 \implies \lambda = a \pm b.$$

The eigenpairs are

$$\lambda_1 = a - b < 0, \quad u_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}; \qquad \lambda_2 = a + b, \quad u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The general solution is

$$y(t) = c_1 e^{(a-b)t} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{(a+b)t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

If $\lambda_2 = a + b < 0$, i.e. $|a| > |b|$, both eigenvalues are negative and the origin is asymptotically stable: both feelings decay to zero. This corresponds to excessive caution ($|a|$ large) overpowering the willingness to open up ($|b|$ small). If $\lambda_2 > 0$, i.e. $|b| > |a|$, the relationship grows without bound along the u_2 direction.

6.4.4 Quick Classification Guide for Phase Portraits

For a 2×2 linear system $y' = Ay$ with $\det(A) \neq 0$, the phase portrait is determined entirely by the eigenvalues λ_1, λ_2 of A. The following guide provides a rapid way to match systems to phase portraits by examining the eigenvalue structure.

Step 1: Calculate the eigenvalues. Find λ_1, λ_2 from the characteristic polynomial $\det(A - \lambda I) = 0$.

Step 2: Classify by type and sign.

- Both real, same sign, both negative: Stable node. All orbits approach the origin. Arrows point inward toward the equilibrium.
- Both real, same sign, both positive: Unstable node. All orbits move away from the origin. Arrows point outward.
- Real, opposite signs: Saddle point. Hyperbolic structure: orbits approach along one eigenvector direction (stable manifold) and recede along the other (unstable manifold). The phase portrait resembles a stretched X.
- Complex with negative real part: Stable (asymptotically stable) spiral. Orbits spiral inward toward the origin. Arrows rotate and shrink.
- Complex with positive real part: Unstable spiral. Orbits spiral outward away from the origin. Arrows rotate and expand.
- Purely imaginary (real part = 0): Center. Orbits trace closed ellipses. Arrows rotate at constant radius; no convergence or divergence.

Remark (Degenerate case: $\det(A) = 0$)

If $\det(A) = 0$, then at least one eigenvalue is 0, and the above classification no longer applies.

- $\lambda_1 = \lambda_2 = 0$, $A = 0$: Every point is an equilibrium. Type: completely degenerate (all equilibria).. Stability: stable (but not asymptotically stable).
- $\lambda_1 = \lambda_2 = 0$, $A \neq 0$: Degenerate equilibrium (defective case). Trajectories are straight lines (typically parallel). Type: degenerate node (non-isolated equilibrium behavior). Stability: unstable.
- One eigenvalue 0, the other $\lambda \neq 0$: The origin is non-isolated; there is a line of equilibria.
  - If $\lambda < 0$: Type: line of equilibria (degenerate node-like). Stability: stable (but not asymptotically stable).
  - If $\lambda > 0$: Type: line of equilibria. Stability: unstable.

Example. *If $\sigma(A) = \{-2, 3\}$ (real, opposite signs), look for a phase portrait with a hyperbolic saddle structure. If $\sigma(A) = \{-1 + 2i, -1 - 2i\}$ (complex with $\mathrm{Re}(\lambda) = -1 < 0$), look for a stable spiral spiraling inward.*

6.5 Stability Theory for Linear Systems

Let (3) hold with $A \in M_n(\mathbb{R})$ and $\det(A) \neq 0$, so that $\tilde{y} = 0 \in \mathbb{R}^n$ is the only equilibrium solution.

Theorem 18 (Stability via Eigenvalues)**.** *Let $A \in M_n(\mathbb{R})$ be invertible.*

1. *If $\mathrm{Re}(\lambda) < 0$ for all $\lambda \in \sigma(A)$, then 0 is an asymptotically stable equilibrium solution of (3).*
2. *If there exists $\lambda_0 \in \sigma(A)$ with $\mathrm{Re}(\lambda_0) > 0$, then 0 is unstable.*

Proof. **(1)** Under this hypothesis, there exist constants $C > 0$ and $\gamma > 0$ such that $\|e^{At}\| \leq Ce^{-\gamma t}$. The unique solution of (3) with $y(0) = y_0$ is $y(t) = e^{At}y_0$, so

$$\|y(t)\| = \|e^{At}y_0\| \leq \|e^{At}\| \cdot \|y_0\| \leq Ce^{-\gamma t}\|y_0\|.$$

Stability: given $\epsilon > 0$, choose $\delta = \epsilon/C$. Then $\|y_0\| < \delta$ implies $\|y(t)\| \leq Ce^{-\gamma t} \cdot \frac{\epsilon}{C} \leq \epsilon$ for all $t \geq 0$.
Asymptotic stability: for any $\alpha > 0$, if $\|y_0\| \leq \alpha$ then $\|y(t)\| \leq C\alpha e^{-\gamma t} \to 0$ as $t \to +\infty$.

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(2) Let $u \pm iv$ be the eigenvector associated to $\lambda_0 = \alpha_0 + i\beta_0$ with $\alpha_0 > 0$. Then

$$y(t) = Ce^{\alpha_0 t}(\cos(\beta_0 t)\, u - \sin(\beta_0 t)\, v)$$

is a solution. Choose $C = \frac{\delta}{\|u\|}$ for any small $\delta > 0$, so that

$$\|y(0)\| = \left\| \frac{\delta}{\|u\|} u \right\| = \delta.$$

Let $\epsilon = 1$. Since $\alpha_0 > 0$, there exists $T = T(\delta)$ such that $\|y(T)\| \geq \epsilon = 1$, so 0 is unstable. $\square$

6.6 Stability Theory for Nonlinear Systems

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Consider a nonlinear system $y' = f(y)$ with $f : D \to \mathbb{R}^n$, $D \subseteq \mathbb{R}^n$ open, and let $\tilde{y} \in D$ be an equilibrium solution, i.e. $f(\tilde{y}) = 0$. For y close to $\tilde{y}$, expand f around $\tilde{y}$:

$$f(y) = f(\tilde{y}) + Df(\tilde{y})(y - \tilde{y}) + O(\|y - \tilde{y}\|^2) \approx Df(\tilde{y})(y - \tilde{y}).$$

Setting $z = y - \tilde{y}$ (translating $\tilde{y}$ to the origin), we get

$$z' = y' = f(y) \approx Df(\tilde{y})z.$$

The solutions of $y' = f(y)$ near $\tilde{y}$ are therefore approximated by the solutions of the *linearization*

$$z' = Df(\tilde{y})z$$

near 0.

Theorem 19 (Stability via Linearization)**.** *Let $f : D \to \mathbb{R}^n$ be C^1 and let $\tilde{y} \in D$ be an equilibrium solution.*

1. *If $\mathrm{Re}(\lambda) < 0$ for all $\lambda \in \sigma(Df(\tilde{y}))$, then $\tilde{y}$ is asymptotically stable.*
2. *If there exists $\lambda_0 \in \sigma(Df(\tilde{y}))$ with $\mathrm{Re}(\lambda_0) > 0$, then $\tilde{y}$ is unstable.*

Definition 45 (Hyperbolic Equilibrium)**.** *An equilibrium solution $\tilde{y}$ is called* hyperbolic *if $\mathrm{Re}(\lambda) \neq 0$ for all $\lambda \in \sigma(Df(\tilde{y}))$.*

Theorem 20 (Hartman–Grobman)**.** *Near a hyperbolic equilibrium $\tilde{y}$, the solutions of $y' = f(y)$ have the same qualitative structure as the solutions of the linearization $z' = Df(\tilde{y})z$ near 0.*

Example. *Let*

$$f(y) = \begin{pmatrix} -2y_1 - y_1^3 - y_1 y_2^2 \\ -3y_2 + y_1^2 y_2 + y_2^3 \end{pmatrix}.$$

Since $f\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, the origin is an equilibrium. The Jacobian is

$$Df(y) = \begin{pmatrix} -2 - 3y_1^2 - y_2^2 & -2y_1 y_2 \\ 2y_1 y_2 & -3 + y_1^2 + 3y_2^2 \end{pmatrix},$$

so

$$Df(0,0) = \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}, \quad \sigma(Df(0,0)) = \{-2, -3\}.$$

Since $\mathrm{Re}(\lambda) < 0$ for all $\lambda \in \sigma(Df(0,0))$, the equilibrium $\tilde{y} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is hyperbolic and asymptotically stable.

7 Laplace Transform

7.1 Definition and Existence

Definition 46 (Laplace Transform)**.** *Given $f : [0, \infty) \to \mathbb{R}$, its Laplace transform is*

$$\mathcal{L}[f](s) = F(s) = \int_0^\infty e^{-st} f(t)\, dt = \lim_{A \to +\infty} \int_0^A e^{-st} f(t)\, dt,$$

provided the improper integral converges.

Definition 47 (Piecewise Continuous)**.** *A function $f : [0, \infty) \to \mathbb{R}$ is piecewise continuous if there exists a partition $0 = t_0 < t_1 < \cdots < t_n < t_{n+1} = +\infty$ such that f is continuous on each (t_i, t_{i+1}) and the one-sided limits $\lim_{t \to t_i^-} f(t)$ and $\lim_{t \to t_i^+} f(t)$ are finite for each i.*

Definition 48 (Exponential Order)**.** *A function $f : [0, \infty) \to \mathbb{R}$ is of exponential order a if there exist constants $K \geq 0$ and $M \geq 0$ such that*

$$|f(t)| \leq K e^{at} \quad \text{for all } t \geq M.$$

Remark 15. *We abbreviate "piecewise continuous of exponential order" as PCEO.*

Example (Exponential order)**.**

- *$f(t) = t^n$ is of exponential order a for any $a > 0$, since t^n grows slower than any exponential.*
- *$f(t) = e^{t^2}$ is not of exponential order for any a.*

Lemma 21. *If $\int_0^\infty |g(t)|\, dt$ converges, then $\int_0^\infty g(t)\, dt$ converges.*

Theorem 22 (Existence of the Laplace Transform)**.** *If f is PCEO with order a, then $F(s) = \mathcal{L}[f](s)$ exists for all $s > a$.*

Proof.

$$\int_0^\infty |e^{-st} f(t)|\, dt \leq \int_0^M e^{-st} |f(t)|\, dt + K \int_M^\infty e^{-(s-a)t}\, dt.$$

The first term is finite since f is piecewise continuous on $[0, M]$. The second term equals

$$K \cdot \frac{e^{-(s-a)M}}{s-a} < \infty \quad \text{for } s > a. \qquad \square$$

7.2 Computing Laplace Transforms

Example ($f(t) = K$)**.** *Since $|K| \leq K\, e^{0 \cdot t}$, the exponential order is $a = 0$. For $s > 0$:*

$$\mathcal{L}[K](s) = K \lim_{A \to \infty} \int_0^A e^{-st}\, dt = K \lim_{A \to \infty} \left(\frac{1}{s} - \frac{e^{-sA}}{s} \right) = \frac{K}{s}.$$

Example ($f(t) = e^{at}$, $a \in \mathbb{R}$)**.** *With $K = 1$, $M = 0$:*

$$\mathcal{L}[e^{at}](s) = \int_0^\infty e^{-(s-a)t}\,dt = \frac{1}{s-a}, \quad s > a.$$

Remark 16. *By induction:* $\mathcal{L}[t^n](s) = \dfrac{n!}{s^{n+1}}$ *for* $n = 0, 1, 2, \ldots$

Example ($f(t) = \cosh(at)$)**.** *Using* $\cosh(at) = \frac{1}{2}(e^{at} + e^{-at})$ *and linearity:*

$$\mathcal{L}[\cosh(at)](s) = \frac{1}{2} \cdot \frac{1}{s-a} + \frac{1}{2} \cdot \frac{1}{s+a} = \frac{s}{s^2 - a^2}, \quad s > |a|.$$

Example ($f(t) = \cos(\omega t)$)**.** *Using* $\cos(\omega t) = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$ *and linearity:*

$$\mathcal{L}[\cos(\omega t)](s) = \frac{1}{2} \cdot \frac{1}{s-i\omega} + \frac{1}{2} \cdot \frac{1}{s+i\omega} = \frac{s}{s^2 + \omega^2}, \quad s > 0.$$

Example (Piecewise function)**.** *Let* $f(t) = \begin{cases} 1, & 0 \le t < 1, \\ K, & t = 1, \\ 0, & t > 1. \end{cases}$ *Since* $f(t) = 0$ *for* $t > 1$*:*

$$\mathcal{L}[f](s) = \int_0^1 e^{-st}\,dt = \begin{cases} 1, & s = 0, \\ \dfrac{1 - e^{-s}}{s}, & s \neq 0, \end{cases}$$

which exists for all $s \in \mathbb{R}$*.*

$f(t)$	$F(s) = \mathcal{L}[f](s)$	Domain
1	$1/s$	$s > 0$
t^n	$n!/s^{n+1}$	$s > 0$
e^{at}	$1/(s-a)$	$s > a$
$\cos(\omega t)$	$s/(s^2+\omega^2)$	$s > 0$
$\sin(\omega t)$	$\omega/(s^2+\omega^2)$	$s > 0$
$\cosh(at)$	$s/(s^2-a^2)$	$s > \lvert a\rvert$
$\sinh(at)$	$a/(s^2-a^2)$	$s > \lvert a\rvert$

7.3 Properties

7.3.1 Linearity

Let f_1, f_2 be PCEO with orders a_1, a_2 and set $a = \max(a_1, a_2)$. Then for any constants c_1, c_2:

$$\mathcal{L}[c_1 f_1 + c_2 f_2](s) = c_1\,\mathcal{L}[f_1](s) + c_2\,\mathcal{L}[f_2](s), \quad s > a.$$

7.3.2 Differentiability and the s-Derivative Formula

Proposition 10. *Let f be PCEO with order a. Then $F : (a, \infty) \to \mathbb{R}$ is C^∞ and*

$$\frac{d^k F}{ds^k}(s) = (-1)^k \, \mathcal{L}[t^k f(t)](s), \quad s > a.$$

Proof. For $k = 1$: since f is PCEO of order a, so is $tf(t)$, so $\mathcal{L}[tf(t)](s)$ exists for $s > a$. Differentiating under the integral sign:

$$F'(s) = \frac{d}{ds} \int_0^\infty e^{-st} f(t) \, dt = -\int_0^\infty te^{-st} f(t) \, dt = -\mathcal{L}[tf(t)](s).$$

For $k > 1$, replace $f(t)$ by $t^{k-1} f(t)$ and iterate. □

7.3.3 Injectivity and the Inverse Transform

Proposition 11 (Injectivity)**.** *If $f, g : [0, \infty) \to \mathbb{R}$ are continuous and of exponential order, and $\mathcal{L}[f] = \mathcal{L}[g]$, then $f = g$.*

Injectivity guarantees that the inverse Laplace transform $\mathcal{L}^{-1}[F(s)]$ is well-defined.

7.3.4 Laplace Transform of Derivatives

Theorem 23. *If f is differentiable and PCEO, then*

$$\mathcal{L}[f'](s) = s \, \mathcal{L}[f](s) - f(0), \tag{11}$$

$$\mathcal{L}[f''](s) = s^2 \, \mathcal{L}[f](s) - sf(0) - f'(0). \tag{12}$$

Proof. Integrating by parts:

$$\mathcal{L}[f'](s) = \int_0^\infty e^{-st} f'(t) \, dt = \Big[e^{-st} f(t)\Big]_0^{+\infty} + s \int_0^\infty e^{-st} f(t) \, dt = -f(0) + s \, \mathcal{L}[f](s).$$

Applying this to f':

$$\mathcal{L}[f''](s) = s \, \mathcal{L}[f'](s) - f'(0) = s\big(s \, \mathcal{L}[f](s) - f(0)\big) - f'(0) = s^2 \mathcal{L}[f](s) - sf(0) - f'(0).$$

□

Example (Solving an IVP via Laplace transform)**.** *Solve $2y'' + 5y' - 3y = 2$, with $y(0) = 0$, $y'(0) = 1$.*

Applying $\mathcal{L}$ to both sides and denoting $Y(s) = \mathcal{L}[y](s)$:

$$\mathcal{L}[y'] = sY(s), \qquad \mathcal{L}[y''] = s^2 Y(s) - 1.$$

The equation becomes

$$2(s^2 Y - 1) + 5(sY) - 3Y = \frac{2}{s},$$

$$(2s^2 + 5s - 3)\,Y(s) = \frac{2}{s} + 2 = \frac{2(s+1)}{s}.$$

Factoring $2s^2 + 5s - 3 = (2s-1)(s+3)$*:*

$$Y(s) = \frac{2(s+1)}{s(2s-1)(s+3)} = \frac{s+1}{s(s-\frac{1}{2})(s+3)}.$$

By partial fractions:

$$Y(s) = -\frac{2}{3}\cdot\frac{1}{s} + \frac{6}{7}\cdot\frac{1}{s-\frac{1}{2}} - \frac{4}{21}\cdot\frac{1}{s+3}.$$

Taking the inverse Laplace transform:

$$\boxed{y(t) = -\frac{2}{3} + \frac{6}{7}\,e^{t/2} - \frac{4}{21}\,e^{-3t}.}$$

7.4 Computational Methods

7.4.1 Integration in the Time Domain

Corollary 23.1. *Let f be PCEO with $F(s) = \mathcal{L}[f](s)$. Then*

$$\mathcal{L}\left[\int_0^t f(u)\,du\right] = \frac{F(s)}{s}, \qquad \textit{equivalently,} \qquad \mathcal{L}^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(u)\,du.$$

Proof. Let $g(t) = \int_0^t f(u)\,du$. Then $g(0) = 0$ and $g'(t) = f(t)$. By the derivative theorem:

$$\mathcal{L}[f](s) = \mathcal{L}[g'](s) = s\,\mathcal{L}[g](s) - g(0) = s\,\mathcal{L}[g](s).$$

Dividing by s gives the result. □

Example. *Compute* $\mathcal{L}^{-1}\left[\frac{1}{s(s^2+1)}\right]$.

Write $\frac{1}{s(s^2+1)} = \frac{1}{s}\cdot\frac{1}{s^2+1} = \frac{F(s)}{s}$ *where* $F(s) = \frac{1}{s^2+1}$*, so* $f(t) = \sin t$*. By Corollary 23.1:*

$$\mathcal{L}^{-1}\left[\frac{1}{s(s^2+1)}\right] = \int_0^t \sin u\,du = \Big[-\cos u\Big]_0^t = 1 - \cos t.$$

Alternatively, by partial fractions: $\frac{1}{s(s^2+1)} = \frac{1}{s} - \frac{s}{s^2+1}$*, giving the same result.*

7.4.2 Convolution

Definition 49 (Convolution)**.** *The convolution of $f, g : [0, \infty) \to \mathbb{R}$ is*

$$(f \star g)(t) = \int_0^t f(t-\tau)\, g(\tau)\, d\tau = \int_0^t f(\xi)\, g(t-\xi)\, d\xi.$$

Theorem 24 (Convolution Theorem)**.** *If f and g are PCEO with $F(s) = \mathcal{L}[f]$ and $G(s) = \mathcal{L}[g]$, then*

$$\mathcal{L}[f \star g](s) = F(s)\, G(s), \qquad \textit{equivalently,} \qquad \mathcal{L}^{-1}[F(s)\, G(s)] = (f \star g)(t).$$

Proof.

$$F(s)\, G(s) = \left(\int_0^\infty f(\xi)\, e^{-s\xi}\, d\xi\right)\left(\int_0^\infty g(\tau)\, e^{-s\tau}\, d\tau\right) = \iint_{[0,\infty)^2} f(\xi)\, g(\tau)\, e^{-s(\xi+\tau)}\, d\xi\, d\tau.$$

Substituting $t = \xi + \tau$ and switching the order of integration:

$$= \int_0^\infty e^{-st} \left(\int_0^t f(t-\tau)\, g(\tau)\, d\tau\right) dt = \mathcal{L}[f \star g](s). \qquad \square$$

Example. *Compute* $\mathcal{L}^{-1}\left[\dfrac{a}{s^2(s^2+a^2)}\right]$.

Write $\dfrac{a}{s^2(s^2+a^2)} = \dfrac{1}{s^2} \cdot \dfrac{a}{s^2+a^2}$. *Since* $\mathcal{L}^{-1}[1/s^2] = t$ *and* $\mathcal{L}^{-1}[a/(s^2 + a^2)] = \sin(at)$*:*

$$\mathcal{L}^{-1}\left[\frac{a}{s^2(s^2+a^2)}\right] = t \star \sin(at) = \int_0^t (t-\tau)\sin(a\tau)\, d\tau = \frac{at - \sin(at)}{a^2}.$$

Example (Integro-differential equation)**.** *Solve* $y'(t) + 2y(t) + \displaystyle\int_0^t y(\tau)\, d\tau = \sin t$, *with* $y(0) = 1$.

Note that $\int_0^t y(\tau)\, d\tau = (1 \star y)(t)$, *so* $\mathcal{L}[1 \star y] = Y(s)/s$. *Let* $Y(s) = \mathcal{L}[y](s)$ *and take the transform:*

$$sY(s) - 1 + 2Y(s) + \frac{Y(s)}{s} = \frac{1}{s^2+1}.$$

Multiplying by s:

$$(s^2 + 2s + 1)\, Y(s) = s + \frac{s}{s^2+1}, \qquad \Longrightarrow \qquad (s+1)^2\, Y(s) = \frac{s(s^2+2)}{s^2+1},$$

$$Y(s) = \frac{s(s^2+2)}{(s+1)^2(s^2+1)}.$$

Using partial fractions and $\mathcal{L}[te^{-t}](s) = 1/(s+1)^2$ *(from the s-derivative formula with* $F(s) = 1/(s+1)$*):*

$$\boxed{y(t) = e^{-t} - \tfrac{3}{2}\, te^{-t} + \tfrac{1}{2}\sin t.}$$

7.5 Step Functions and Translation Theorems

Definition 50 (Unit step function / Heaviside function)**.**

$$u_c(t) = \begin{cases} 0, & 0 \le t < c, \\ 1, & t \ge c. \end{cases}$$

Example. *Any piecewise function can be written with step functions:*

$$f(t) = \begin{cases} f_1(t), & 0 \le t < a, \\ f_2(t), & a \le t < b, \\ f_3(t), & t \ge b \end{cases} \quad \Longrightarrow \quad f = u_0\, f_1 + u_a(f_2 - f_1) + u_b(f_3 - f_2).$$

Theorem 25 (Translation Theorem 1)**.**

$$\mathcal{L}[u_c](s) = \int_c^\infty e^{-st}\, dt = \frac{e^{-cs}}{s}, \quad s > 0.$$

Definition 51 (Translation operator)**.**

$$\mathsf{t}_c(f)(t) = \begin{cases} 0, & 0 \le t < c, \\ f(t-c), & t \ge c. \end{cases}$$

This shifts f to the right by c and fills the gap with zero.

Theorem 26 (Translation Theorem 2 – Time Shift)**.** *Let f be PCEO with order a and let $c > 0$. Then*

$$\mathcal{L}[\mathsf{t}_c(f)](s) = e^{-cs}\, \mathcal{L}[f](s), \qquad \textit{equivalently,} \qquad \mathcal{L}^{-1}[e^{-cs}F(s)] = \mathsf{t}_c(f)(t).$$

Proof.

$$\mathcal{L}[\mathsf{t}_c(f)](s) = \int_c^\infty e^{-st} f(t-c)\, dt.$$

Substituting $u = t - c$:

$$= \int_0^\infty e^{-s(u+c)} f(u)\, du = e^{-cs} \int_0^\infty e^{-su} f(u)\, du = e^{-cs}\, \mathcal{L}[f](s). \qquad \square$$

Key identity for TT2: to apply the time-shift theorem, write the function in the form $u(t-c) \cdot g(t-c)$, then

$$\mathcal{L}\{u(t-c)\, g(t-c)\} = e^{-cs}\, \mathcal{L}\{g(t)\}.$$

Theorem 27 (Translation Theorem 3 – Frequency Shift)**.** *Let f be PCEO with order a and let $c \in \mathbb{R}$. Then*

$$\mathcal{L}[e^{ct} f(t)](s) = F(s-c), \quad s > a+c, \qquad \textit{equivalently,} \qquad \mathcal{L}^{-1}[F(s-c)] = e^{ct}\, \mathcal{L}^{-1}[F(s)].$$

Proof.

$$\mathcal{L}[f](s-c) = \int_0^\infty e^{-(s-c)t} f(t)\,dt = \int_0^\infty e^{-st}\left[e^{ct} f(t)\right] dt = \mathcal{L}[e^{ct} f(t)](s). \quad \square$$

Example (TT2: piecewise with cosine)**.** *Let* $f(t) = \begin{cases} \sin t, & 0 \le t < \pi/4, \\ \sin t + \cos(t - \pi/4), & t \ge \pi/4. \end{cases}$

Rewrite: $f(t) = \sin t + u(t - \frac{\pi}{4}) \cos(t - \frac{\pi}{4})$*. By TT2:*

$$\mathcal{L}\{f\} = \frac{1}{s^2+1} + e^{-\frac{\pi}{4}s} \cdot \frac{s}{s^2+1}.$$

Example (TT2: three-piece function)**.** *Let* $f(t) = \begin{cases} 2, & 0 \le t < 1, \\ 1+t, & 1 \le t < 4, \\ t^2, & t \ge 4. \end{cases}$

Write using step functions:

$$f(t) = 2 + u(t-1)(t-1) + u(t-4)(t^2 - t - 1).$$

Set $g_1(t) = t$ *so that* $u(t-1)(t-1) = u(t-1)\, g_1(t-1)$*, and* $g_2(t) = t^2 + 7t + 11$ *so that* $u(t-4)(t^2 - t - 1) = u(t-4)\, g_2(t-4)$*. By TT2:*

$$\mathcal{L}\{f\} = \frac{2}{s} + e^{-s} \cdot \frac{1}{s^2} + e^{-4s}\left(\frac{2}{s^3} + \frac{7}{s^2} + \frac{11}{s}\right).$$

Example (Inverse transform with TT2)**.** *Compute* $\mathcal{L}^{-1}\left\{\frac{1-e^{-2s}}{s^2}\right\}$.

Split: $\frac{1-e^{-2s}}{s^2} = \frac{1}{s^2} - e^{-2s} \cdot \frac{1}{s^2}$*. Since* $\mathcal{L}^{-1}\{1/s^2\} = t$*, by TT2:*

$$f(t) = t - u(t-2)(t-2) = \begin{cases} t, & 0 \le t < 2, \\ 2, & t \ge 2. \end{cases}$$

Example (TT3: completing the square)**.** *Find* $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 4s + 5}\right\}$.

Complete the square: $s^2 - 4s + 5 = (s-2)^2 + 1$*, so*

$$\frac{1}{s^2 - 4s + 5} = \frac{1}{(s-2)^2 + 1} = F(s-2), \quad \textit{where } F(s) = \frac{1}{s^2+1}.$$

Since $\mathcal{L}^{-1}\{F(s)\} = \sin t$*, by TT3:*

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 4s + 5}\right\} = e^{2t} \sin t.$$

7.6 ODEs with Discontinuous Forcing Terms

Example. *Solve* $y'' + 4y = f(t)$, *where* $f(t) = \begin{cases} 1, & 0 \le t < 1, \\ 0, & t \ge 1, \end{cases}$ *with* $y(0) = y'(0) = 0$.

Write $f(t) = u_0(t) - u_1(t)$ *and take the Laplace transform:*

$$s^2 Y(s) + 4Y(s) = \frac{1}{s} - \frac{e^{-s}}{s}.$$

Let $F(s) = \dfrac{1}{s(s^2+4)}$. *By convolution:*

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s} \cdot \frac{1}{s^2+4}\right\} = 1 \star \tfrac{1}{2}\sin(2t) = \int_0^t \tfrac{1}{2}\sin(2\tau)\,d\tau = \tfrac{1}{4}(1-\cos 2t).$$

Denote $g(t) = \frac{1}{4}(1-\cos 2t)$. *By TT2:*

$$y(t) = g(t) - \mathtt{t}_1[g](t) = \begin{cases} \frac{1}{4}(1-\cos 2t), & 0 \le t < 1, \\ \frac{1}{4}(1-\cos 2t) - \frac{1}{4}(1-\cos 2(t-1)), & t \ge 1. \end{cases}$$

7.7 Impulse Functions

Consider the rectangular pulse of width 2τ and unit area:

$$d_\tau(t) = \begin{cases} \dfrac{1}{2\tau}, & |t| < \tau, \\ 0, & |t| \ge \tau. \end{cases} \qquad \text{Note: } \int_{-\infty}^{\infty} d_\tau(t)\,dt = \frac{2\tau}{2\tau} = 1.$$

The impulse function is the limiting "function" $\delta(t) = \lim_{\tau\to 0} d_\tau(t)$. For $t \ne 0$, $\delta(t) = 0$; at $t = 0$, $\delta(t) = \lim_{\tau\to 0} 1/(2\tau) = \infty$. Crucially, $\lim_{\tau\to 0}\int_{-\infty}^{\infty} d_\tau(t)\,dt = 1$.

Definition 52 (Dirac delta function)**.** *The impulse function (Dirac delta function), with impulse at* $t = t_0$, *is defined by:*

1. $\delta(t - t_0) = 0$ *for all* $t \ne t_0$.
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t-t_0)\,dt = 1$.

Theorem 28. *For* $t_0 > 0$*:*

$$\mathcal{L}\{\delta(t-t_0)\} = e^{-st_0}.$$

In particular, $\mathcal{L}\{\delta(t)\} = 1$.

Proof.

$$\begin{aligned} \mathcal{L}\{\delta(t-t_0)\} &= \lim_{\tau\to 0}\mathcal{L}\{d_\tau(t-t_0)\} = \lim_{\tau\to 0}\int_{t_0-\tau}^{t_0+\tau} e^{-st}\cdot\frac{1}{2\tau}\,dt \\ &= \lim_{\tau\to 0}\frac{1}{2\tau}\cdot\frac{e^{-s(t_0-\tau)} - e^{-s(t_0+\tau)}}{s} = e^{-st_0}\lim_{\tau\to 0}\frac{e^{s\tau} - e^{-s\tau}}{2s\tau}. \end{aligned}$$

By L'Hôpital's rule:

$$\lim_{\tau\to 0}\frac{e^{s\tau}-e^{-s\tau}}{2s\tau}=\lim_{\tau\to 0}\frac{s(e^{s\tau}+e^{-s\tau})}{2s}=1.$$

Therefore $\mathcal{L}\{\delta(t-t_0)\}=e^{-st_0}$. □

Example. *Solve $y''+2y'+y=\delta(t-5)$, with $y(0)=y'(0)=0$.*

Taking the Laplace transform:

$$(s^2+2s+1)\,Y(s)=e^{-5s}\quad\Longrightarrow\quad Y(s)=\frac{e^{-5s}}{(s+1)^2}.$$

By TT2: $y(t)=\mathsf{t}_5\left[\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}\right]$.

By TT3: $\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}=e^{-t}\,\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}=te^{-t}$.

Therefore:

$$\boxed{y(t)=u_5(t)\,(t-5)\,e^{-(t-5)}=\begin{cases}0, & 0\le t<5,\\ (t-5)\,e^{-(t-5)}, & t\ge 5.\end{cases}}$$

8 Power Series Solutions and Numerical Methods

Remark 17. *Make sure to read Section 6.1 from the lecture notes.*

8.1 Second-Order Non-Autonomous Linear ODEs

Consider the general equation

$$P(x)\,y''(x)+Q(x)\,y'(x)+R(x)\,y(x)=0. \tag{13}$$

Example (Important special cases)**.**

- *Bessel equation:* $x^2y''+xy'+(x^2-\nu^2)y=0$.
- *Legendre equation:* $(1-x^2)y''-2xy'+\alpha(\alpha+1)y=0$.
- *Chebyshev equation:* $(1-x^2)y''-xy'+\alpha^2y=0$.

Definition 53 (Ordinary point)**.** *A point $x=x_0$ is an ordinary point of (13) if $P(x_0)\neq 0$.*

Near an ordinary point, $P\neq 0$, so we can divide through to get the standard form

$$y''+p(x)\,y'+q(x)\,y=0,\qquad p(x)=\frac{Q(x)}{P(x)},\quad q(x)=\frac{R(x)}{P(x)}. \tag{14}$$

Key assumption: p and q are analytic at x_0, i.e. they have power series expansions

$$p(x) = \sum_{n\geq 0} p_n (x - x_0)^n, \qquad q(x) = \sum_{n\geq 0} q_n (x - x_0)^n$$

with positive radii of convergence. By the Cauchy–Kovalevskaya theorem (originally for PDEs), if p and q are analytic at x_0, then $y(x)$ is analytic at $x = x_0$.

8.2 The Power Series Method

Since the solution is analytic, write $y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$, with $a_0 = y(x_0)$ and $a_1 = y'(x_0)$. The derivatives are:

$$y'(x) = \sum_{n=0}^{\infty} (n+1)\, a_{n+1} (x - x_0)^n,$$
$$y''(x) = \sum_{n=0}^{\infty} (n+2)(n+1)\, a_{n+2} (x - x_0)^n.$$

The products $p(x)y'(x)$ and $q(x)y(x)$ are computed via the Cauchy product:

$$p(x)y'(x) = \sum_{n\geq 0} (p \star \tilde{a})_n \, (x - x_0)^n, \qquad q(x)y(x) = \sum_{n\geq 0} (q \star a)_n \, (x - x_0)^n,$$

where $\tilde{a}_k = (k+1)a_{k+1}$ and $(p \star \tilde{a})_n = \sum_{k=0}^{n} (k+1)\, a_{k+1}\, p_{n-k}$.

Substituting into (14) and matching coefficients gives the recurrence relation:

$$\boxed{a_{n+2} = \frac{-1}{(n+2)(n+1)} \Big[(p \star \tilde{a})_n + (q \star a)_n \Big], \quad n \geq 0.}$$

Example ($y'' + y = 0$)**.** *Here $p = 0$, $q = 1$, $x_0 = 0$. The recurrence simplifies to:*

$$(n+2)(n+1)\, a_{n+2} + a_n = 0 \quad \Longrightarrow \quad a_{n+2} = -\frac{a_n}{(n+2)(n+1)}.$$

With a_0 and a_1 free, this produces the familiar series for cos *and* sin.

Example (Airy's equation: $y'' - xy = 0$, $y(0) = 1$, $y'(0) = 0$)**.** *Take $x_0 = 0$. Substituting:*

$$y''(x) = \sum_{n=0}^{\infty} (n+2)(n+1)\, a_{n+2}\, x^n, \qquad xy(x) = \sum_{n=1}^{\infty} a_{n-1}\, x^n.$$

Setting $y'' - xy = 0$:

$$2a_2 + \sum_{n=1}^{\infty} \Big[(n+2)(n+1)\, a_{n+2} - a_{n-1} \Big] x^n = 0.$$

This gives $a_2 = 0$ and the recurrence:

$$a_{n+2} = \frac{a_{n-1}}{(n+2)(n+1)}, \quad n \geq 1.$$

With $a_0 = 1$, $a_1 = 0$: non-zero coefficients occur at indices $n = 0, 3, 6, 9, \ldots$ The Airy function is:

$$y(x) = \sum_{k=0}^{\infty} \frac{x^{3k}}{(2 \cdot 3)(5 \cdot 6) \cdots (3k-1)(3k)}.$$

By the ratio test:

$$\left| \frac{b_{k+1}}{b_k} \right| = \frac{|x|^3}{(3k+2)(3k+3)} \to 0 \quad \text{as } k \to \infty,$$

so the radius of convergence is $\rho = +\infty$.

Problem 5. *Redo Airy's equation with $y(0) = 0$, $y'(0) = 1$ (this gives the second linearly independent solution).*

8.3 Power Series for IVPs in Nonlinear Systems

Consider the IVP

$$\dot{y} = f(y), \qquad y(0) = y_0 \in U \subseteq \mathbb{R}^n,$$

where $f : U \to \mathbb{R}^n$ is analytic and U is open. By the Cauchy–Kovalevskaya theorem, the solution is analytic, so we seek $y(t) = \sum_{n=0}^{\infty} a_n t^n$ with $a_0 = y_0$.

Substituting into the ODE:

$$\sum_{n=0}^{\infty} (n+1)\, a_{n+1}\, t^n = f\left(\sum_{n=0}^{\infty} a_n t^n \right) = \sum_{n=0}^{\infty} \Phi_n(a)\, t^n,$$

giving the recurrence:

$$\boxed{a_{n+1} = \frac{\Phi_n(a)}{n+1}, \quad n \geq 0, \qquad a_0 = y_0.}$$

Example (Logistic equation: $y' = y(1-y)$, $y(0) = 1/2$)**.** *Here $f(y) = y - y^2$. Expanding y^2 via the Cauchy product:*

$$y^2(t) = \sum_{n=0}^{\infty} (a \star a)_n\, t^n, \qquad (a \star a)_n = \sum_{k=0}^{n} a_k\, a_{n-k}.$$

The recurrence becomes:

$$a_{n+1} = \frac{1}{n+1} \Big[a_n - (a \star a)_n \Big], \quad n \geq 0.$$

With $a_0 = 1/2$: $a_1 = 1/4$, $a_2 = 0$, $a_3 = -1/48$, $a_4 = 0$, $a_5 = 1/480$, so

$$y(t) = \frac{1}{2} + \frac{1}{4}\, t - \frac{1}{48}\, t^3 + \frac{1}{480}\, t^5 + \cdots \approx \sum_{n=0}^{N} a_n t^n.$$

8.4 Numerical Method via Power Series

Given $y' = f(y)$, $y(0) = y_0$, fix an order N and compute $a_0, a_1, \ldots, a_N$ from the recurrence. This gives the numerical approximation of order N:

$$y^N(t) = \sum_{n=0}^{N} a_n\, t^n.$$

Case $N = 1$ (Euler's method):

$$y^1(t) = a_0 + a_1 t = y_0 + f(y_0)\, t.$$

Picking a step size $h > 0$ and iterating:

$$\boxed{y_{n+1} = y_n + h\, f(y_n), \qquad t_{n+1} = t_n + h.}$$

Case $N > 1$ (Taylor method of order N): compute $y_1 = \sum_{n=0}^{N} a_n h^n$ and apply the method iteratively, achieving higher-order accuracy.

9 Appendix

9.1 Picard Iterations Convergence

Proof of Picard Iterations Convergence. Pick $t \in [t_0, t_0 + \epsilon]$ (the proof is similar for $t \in [t_0 - \epsilon, t_0]$). The goal is to show that $\{y_k(t)\}_{k=0}^{\infty}$ is a Cauchy sequence[7] in $\mathbb{R}^n$.[8] We prove by induction that

$$(**) \quad ||y_m(t) - y_{m-1}(t)|| \leq L^{m-1} M_{\alpha,\delta} \frac{(t - t_0)^m}{m!}, \ \forall m \geq 1.$$

Base case $m = 1$:

$$||y_1(t) - y_0(t)|| = \left\| \int_{t_0}^{t} f(y_0, s) ds \right\| \leq \int_{t_0}^{t} ||f(y_0, s)|| ds \leq M_{\alpha,\delta} |t - t_0| \leq \alpha.$$

Induction step:

$$||y_{m+1}(t) - y_m(t)|| \leq \int_{t_0}^{t} ||f(y_m(s), s) - f(y_{m-1}(s), s)|| ds.$$

Since $D_{\alpha,\delta}$ is compact and f is Lipschitz on $D_{\alpha,\delta}$, there exists L such that

$$||f(x, t) - f(y, t)|| \leq L||x - y||.$$

Thus,

$$\leq L \int_{t_0}^{t} ||y_m(s) - y_{m-1}(s)|| ds.$$

[7] Cauchy sequence is a sequence that has a limit in a metric space $\mathbb{R}^n$.

[8] Evaluated on compacted cylinder, understanding epsilon, the L Lipschitz constant coming from somewhere, not on the analysis background such as the Banach fixed point theorem.

Using (**),

$$\leq L^m M_{\alpha,\delta} \frac{1}{(m-1)!} \int_{t_0}^{t} (s-t_0)^{m-1} ds = L^m M_{\alpha,\delta} \frac{(t-t_0)^m}{m!}.$$

Hence (**) holds. In particular, for all $\rho \geq 1$,

$$||y_\rho(t) - y_{\rho-1}(t)|| \leq M_{\alpha,\delta} \frac{(L(t-t_0))^\rho}{(\rho)!} < \frac{M_{\alpha,\delta}}{L} \frac{(L\epsilon)^\rho}{\rho!}.$$

Let $m, p \geq 1$:

$$||y_{m+p}(t) - y_{m+1}(t)|| \leq \sum_{k=1}^{p-1} ||y_{m+k+1}(t) - y_{m+k}(t)||.$$

So,

$$< \frac{M_{\alpha,\delta}}{L} \sum_{j=m+2}^{m+p} \frac{(L\epsilon)^j}{j!}.$$

Since $e^{L\epsilon} = \sum_{j=0}^{\infty} \frac{(L\epsilon)^j}{j!}$ converges,

$$\to_{m,p\to+\infty} 0.$$

Thus $\{y_k(t)\}$ is Cauchy and converges to $y(t)$. Taking limits in the iteration,

$$y(t) = y_0 + \int_{t_0}^{t} f(y(s), s) ds.$$

□

10 Solutions

Solution 1 (System of First Order ODEs)**.** *Define new variables:*

$$y_1 = y, \quad y_2 = y', \quad y_3 = y''.$$

Now we can rewrite the third order ODE into a first order system:

$$\begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ y_3' = -4y_2 + y_1 \end{cases}$$

Solution 2 (Existence and Uniqueness Theorem)**.** *Let $f(y,t) = t^2 + y^2$. This function is continuous and has continuous partial derivatives $\forall (t,y) \in \mathbb{R}^2$, ensuring a unique local solution. To find the guaranteed interval of existence ϵ, we define a rectangular domain D centered at the origin:*

$$D_{\alpha,\delta} = \{(y,t) : \|y\| \leq \alpha, |t| \leq \delta\}.$$

Now we need to bound the function. We find the maximum value M of $|f(t,y)|$ on this domain:

$$M = \sup_{(t,y)\in D_{\alpha,\delta}} |t^2 + y^2| = \delta^2 + \alpha^2.$$

Now we determine the Existence Interval. According to the theorem, the solution is guaranteed to exist for $|t| \leq \epsilon$, where:

$$\epsilon = \min\left(\delta, \frac{\alpha}{M}\right) = \min\left(\delta, \frac{\alpha}{\delta^2 + \alpha^2}\right).$$

To find the largest possible value for ϵ, we first maximize $h(\alpha) = \frac{\alpha}{\delta^2+\alpha^2}$ with respect to α for a fixed δ:

$$h'(\alpha) = \frac{(\delta^2 + \alpha^2)(1) - \alpha(2\alpha)}{(\delta^2 + \alpha^2)^2} = \frac{\delta^2 - \alpha^2}{(\delta^2 + \alpha^2)^2}.$$

Setting $h'(\alpha) = 0$ gives $\alpha = \delta$.Substituting $\alpha = \delta$ back into our expression for ϵ:

$$\epsilon = \min\left(\delta, \frac{\delta}{\delta^2 + \delta^2}\right) = \min\left(\delta, \frac{1}{2\delta}\right).$$

The maximum value of this minimum occurs when the two terms are equal:

$$\delta = \frac{1}{2\delta} \implies \delta^2 = \frac{1}{2} \implies \delta = \frac{1}{\sqrt{2}}.$$

The largest guaranteed interval of existence ϵ using this rectangular domain is:

$$\epsilon = \frac{1}{\sqrt{2}} \approx 0.707.$$

Solution 3 (LLC)**.** *We have*

$$f(y) = 3y^{2/3}, \qquad f'(y) = 2y^{-1/3}.$$

This derivative is not bounded near $y = 0$, so f is not locally Lipschitz at $y = 0$. The function

$$y_1(t) = 0$$

is a solution. In Chapter 2, solving the separable equation (assuming $y \neq 0$) gives another solution

$$y_2(t) = \begin{cases} t^3, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Indeed,

$$y_2'(t) = 3t^2 = 3(y_2(t))^{2/3}.$$

Thus, the solution is not unique.

Solution 4 (Complex eigenvalues and real-valued solutions)**.** *We are given the system*

$$\vec{y}' = A\vec{y}, \qquad A = \begin{pmatrix} -3 & -2 \\ 5 & 3 \end{pmatrix}.$$

(a) Eigenvalues and eigenvectors. *The characteristic polynomial is*

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} -3-\lambda & -2 \\ 5 & 3-\lambda \end{vmatrix} = (-3-\lambda)(3-\lambda) + 10 = \lambda^2 + 1.$$

Setting $p(\lambda) = 0$ gives us the eigenvalues $\lambda_1 = i$ and $\lambda_2 = -i$.

For $\lambda_1 = i$, we solve $(A - iI)u = 0$:

$$\begin{pmatrix} -3-i & -2 \\ 5 & 3-i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0.$$

From the first row, $(-3-i)a - 2b = 0$ implies $a = \frac{2}{-3-i}b$. Choosing $b = 1$, we obtain our first solution

$$u_1 = \begin{pmatrix} \dfrac{2}{-3-i} \\ 1 \end{pmatrix}.$$

For $\lambda_2 = -i$, we similarly solve $(A + iI)u = 0$:

$$\begin{pmatrix} -3+i & -2 \\ 5 & 3+i \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0,$$

which gives $a = \frac{2}{-3+i}b$. Again choosing $b = 1$, we obtain our second solution

$$u_2 = \begin{pmatrix} \dfrac{2}{-3+i} \\ 1 \end{pmatrix}.$$

Hence,

$$\boxed{\lambda_1 = i, \quad \vec{v}_1 = \begin{pmatrix} \dfrac{2}{-3-i} \\ 1 \end{pmatrix}, \qquad \lambda_2 = -i, \quad \vec{v}_2 = \begin{pmatrix} \dfrac{2}{-3+i} \\ 1 \end{pmatrix}.}$$

(b) Real-valued solution of the initial value problem.
From the eigenpair (λ_1, u_1)[9], a complex-valued solution is

$$\vec{y}(t) = e^{it}u_1.$$

Using Euler's formula (6), we write

$$\vec{y}(t) = (\cos t + i \sin t) \begin{pmatrix} \dfrac{2}{-3-i} \\ 1 \end{pmatrix} = (\cos t + i \sin t) \begin{pmatrix} -\dfrac{3}{5} + \dfrac{1}{5}i \\ 1 \end{pmatrix}.$$

[9] Since the matrix has real coefficients, the second eigenpair (λ_2, u_2) is simply the complex conjugate of (λ_1, u_1), so it produces the same two real solutions after taking real and imaginary parts. Therefore, working with one eigenpair is sufficient.

Separating real and imaginary parts yields two real-valued solutions:

$$\vec{y}_1(t) = \begin{pmatrix} -3\cos t - \sin t \\ 5\cos t \end{pmatrix}, \qquad \vec{y}_2(t) = \begin{pmatrix} \cos t - 3\sin t \\ 5\sin t \end{pmatrix}.$$

Therefore, the general real solution is

$$\vec{y}(t) = c_1\vec{y}_1(t) + c_2\vec{y}_2(t).$$

Applying the initial conditions $y_(0) = \begin{pmatrix} 7 \\ -15 \end{pmatrix}$, *we compute*

$$\vec{y}_1(0) = \begin{pmatrix} -3 \\ 5 \end{pmatrix}, \qquad \vec{y}_2(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Thus,

$$\begin{pmatrix} 7 \\ -15 \end{pmatrix} = c_1 \begin{pmatrix} -3 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Solving gives

$$c_1 = -3, \qquad c_2 = -2.$$

Therefore, the real-valued solution of the initial value problem is

$$\vec{y}(t) = c_1\vec{y}_1(t) + c_2\vec{y}_2(t) = -3\begin{pmatrix} -3\cos t - \sin t \\ 5\cos t \end{pmatrix} - 2\begin{pmatrix} -3\sin t + \cos t \\ 5\sin t \end{pmatrix}.$$

You can simplify each component if desired:

$$\vec{y}(t) = \begin{pmatrix} 9\cos t + 3\sin t + 6\sin t - 2\cos t \\ -15\cos t - 10\sin t \end{pmatrix} = \begin{pmatrix} 7\cos t + 9\sin t \\ -15\cos t - 10\sin t \end{pmatrix}.$$

11 Useful Links